Backward-branch negative reflection of guided waves from the free edge of a flexoelectric piezoelectric semiconductor plate

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Abstract

This paper presents a model of the scattering of a guided wave at the free edge of an n-type ZnO plate ($6mm$ symmetry), in which strain-gradient elasticity, piezoelectricity, flexoelectricity and linearized electron drift–diffusion act together, with a carrier–strain-gradient coupling added to the electric enthalpy. The governing equations follow from a variational principle, which also yields the edge conditions, including the double traction and the corner line force. For a wave travelling along the plate they reduce to a quartic eigenvalue problem, which is solved at fixed frequency by a spectral Galerkin expansion through the thickness; the incident mode is then matched to the outgoing modes at the traction-free edge. The partition of the incident power shows that a backward mode, whose energy flux is opposite to its phase velocity, carries 1.93 % of the power at the chosen operating point and up to 94.29 % near a zero-group-velocity point of the same branch. These are modal shares and not reflectances, because the incident flux itself becomes small there. The conversion belongs to the backward branch of the plate and not to the added coupling, which acts mainly through conduction loss and lowers the share, and near the zero-group-velocity point the share is sensitive to carrier density and to elastic loss.

Keywords: flexoelectric semiconductor; strain-gradient elasticity; guided plate waves; backward branch; negative reflection; spectral Galerkin method.

Nomenclature

Symbol	Quantity	Symbol	Quantity
u_i	displacement	v_i	test displacement
ε_{ij}	small strain	$\eta_{ijk} = \varepsilon_{ij,k}$	strain gradient
σ_{ij}	Cauchy stress	τ_{ijk}	hyperstress ($\tau_{ijk} = \tau_{jik}$)
Σ_{ij}	$\sigma_{ij} - \partial_k \tau_{ijk}$	n_i	surface normal
c_{ijkl}	elastic stiffness	A_{ijklmn}	strain-gradient modulus
e_{kij}	piezoelectric constant	f_{ijkl}	flexoelectric constant
κ_{ij}	permittivity	ℓ	strain-gradient length
φ	electrostatic potential	ψ	test potential
$E_i = -\partial_i \varphi$	electric field	D_i	electric displacement
Δn	carrier-number perturbation	$\widehat{N} = \widetilde{\mu} + Q\widehat{\varphi} - h\widehat{u}_3''$	carrier amplitude
n_0	steady carrier density	μ_e	electron mobility
D	electron diffusivity	q	elementary charge
$\widetilde{\mu}$	electrochemical potential	Φ	test electrochemical potential
H_{jkl}, $H \equiv H_{zzz}$	carrier–strain-gradient coupling	H/H^*	normalized coupling
$a = k_B T/n_0$	carrier compressibility	Q	dimensionless charge
k_x	propagation wavenumber	k_y	transverse wavenumber
k_{Bx}	backward-branch wavenumber	k_{ix}	incident wavenumber
d	plate thickness	$\zeta = 2z/d$	scaled thickness
ω, f	angular frequency, frequency	$\lambda = \ell/d$	dimensionless gradient length
$x = 0$	reflecting edge	L	reference-plane distance
R_B	backward modal share	P_x	thickness-integrated power flux
P_{inc}	incident self-power	θ_E	energy angle
$\alpha_{\mathrm{d}} = D/(dc_s)$	transport-over-elastic rate	$\alpha = 2\lvert \operatorname{Im} k_x \rvert$	spatial attenuation

1 Introduction

Piezoelectric semiconductors, such as n-type zinc oxide, combine a piezoelectric response with mobile charge carriers. This combination is important in nanogenerators and piezotronic devices [1], and ZnO is a well-characterized material [2]. Acoustic waves in such media are coupled to the carrier dynamics. The accompanying piezoelectric field drives carrier redistribution, leading to field screening and energy dissipation [3], and drifting carriers can even amplify ultrasound [4]. The linearized drift–diffusion model has been incorporated into the elastic and piezoelectric equations for both rods [5] and plates [6]. The same model is adopted in the present study.

For plates a few hundred nanometers thick operating at frequencies of around ten gigahertz, two additional effects become important. The first is flexoelectricity, the polarization induced by a strain gradient rather than by a strain [7, 8, 9, 10, 11]. Since this effect scales inversely with a characteristic length, it can become comparable to piezoelectricity in thin films and near free surfaces and edges. Its significance in semiconductors has also been demonstrated [12]. The second is the dependence of the elastic energy on the strain gradient, which introduces an internal length into the equations of motion [13]. In semiconductors, these effects interact with the mobile carriers. The bound charge induced by the strain gradient is screened by the carriers, which in turn modify the electric field. This interaction forms the basis of flexoelectric semiconductor theory [14, 15, 16]. A thermodynamic framework for this theory was subsequently developed by Qu and co-workers [17], and a computational treatment of flexoelectric semiconductor junctions has also appeared [18].

Wave propagation in flexoelectric media has been studied for several geometries. Previous studies have examined Lamb waves in plates with flexoelectricity and strain-gradient elasticity [19] and Rayleigh waves on a flexoelectric half-space [20]. Reflection and transmission have also been investigated for a flexoelectric slab [21], for a piezoelectric semiconductor slab

with carrier and bias effects [22], and at the free surface of a pre-stressed flexoelectric half-space [23]. For flexoelectric and piezoelectric semiconductors, Rayleigh waves [24, 25] and the exact multi-field solution for a plate [6] have been analyzed, and guided waves in plates with strain-gradient elasticity alone are available for the Mindlin Form-II and generalized first strain-gradient theories [26, 27]. These studies mainly address wave dispersion and bulk-wave reflection at half-space boundaries.

The scattering of a guided mode at a free plate edge is, by contrast, a classical problem for purely elastic plates. The edge couples the propagating Lamb modes to evanescent modes, and the partition of the incident power among the propagating modes was obtained by modal expansion and variational methods [28, 29, 30], compared with measurement [31] and extended to oblique incidence [32]. The same edge also supports a resonance at frequencies of zero group velocity [33], where the dispersion branch has a turning point. Lamb modes also include backward modes, for which the energy flux is opposed to the phase velocity [34]. When forward and backward modes coexist near a zero-group-velocity point [35], the free edge can convert one into the other, producing negative reflection [36, 37, 38]. A similar mode-conversion mechanism at thickness steps can produce negative refraction [39], analogous to negative refraction in elastic metamaterials [40]. Semi-analytical methods are widely used to study guided waves in plates. These methods discretize the plate thickness while retaining an analytical description of the propagation direction [41, 42]. The Legendre–Galerkin construction [43] also provides well-conditioned systems for the fourth-order operators arising in strain-gradient elasticity.

Despite these developments, an important gap remains in the understanding of guided-wave scattering at the free edge of flexoelectric semiconductor plates. The guided-mode edge-scattering studies cited above consider purely elastic plates and do not account for strain-gradient elasticity, flexoelectricity or mobile carriers. Existing flexoelectric-semiconductor studies have focused on modal dispersion and attenuation or on body-wave reflection at half-spaces. They do not show how an incident guided mode is distributed among the outgoing modes at a free plate edge. In a strain-gradient plate the discrete spectrum may include backward modes, while carrier-induced attenuation can modify the power distribution among the outgoing branches. Unlike dispersion analysis, edge scattering requires the outgoing modes to satisfy the traction-free conditions at the edge. For a strain-gradient plate these conditions include both the generalized traction and the independent double traction.

The physical significance of this problem becomes particularly clear near a zero-group-velocity point. Negative reflection in elastic plates arises there from the coexistence of a forward and a backward mode, and it is not known whether the conversion survives when an internal length, flexoelectric polarization and mobile carriers change the dispersion and add loss. The low power flux in this regime may make the scattering response sensitive to carrier-induced attenuation. The power distribution among the individual outgoing modes is therefore more informative than the total reflected power alone.

The present study investigates guided-wave scattering at the free edge of an n-type ZnO plate. The model incorporates strain-gradient elasticity, piezoelectricity, flexoelectricity and linearized drift–diffusion, together with one additional carrier–strain-gradient coupling derived from a thermodynamic potential (section 2). The study addresses two questions: whether a backward branch exists in the frequency range considered and how efficiently the free edge converts the incident mode into that branch. It also examines whether the additional carrier–strain-gradient coupling modifies this conversion.

The novelty and the main contributions of this work are as follows.

1. To the authors' knowledge, this is the first investigation of guided-wave scattering at the free edge of a piezoelectric semiconductor plate incorporating both flexoelectricity

and strain-gradient elasticity.
2. The generalized traction and the independent double traction are imposed separately, and the corner line force at the edge–face intersection is also derived and evaluated.
3. The incident power is resolved into the outgoing branches, and the modal share of the backward branch is distinguished from a reflectance.
4. The conversion is traced to the backward-mode branch and its zero-group-velocity fold, whose frequency shifts with the gradient length, and it persists without the additional coupling.
5. Matched control cases quantify the effect of the carrier–strain-gradient coupling and separate it from the conversion due to the backward branch.
6. The solution is validated against an independent C^1 finite-element plate, a two-dimensional finite-element solution, exact limits and an energy balance.

The plate occupies $x < 0$, with the reflecting edge at $x = 0$. The z axis is the six-fold axis of ZnO and points through the thickness, with the faces at $z = \pm d/2$. A mode is called *backward* if its energy flux has a negative x component. An *outgoing* mode either decays away from the edge or, if it propagates, carries energy away from it. The formulation and the solution method are presented first, then the numerical results, and the findings are summarized in the conclusions.

2 Mathematical formulation

The formulation follows the order in which one enthalpy generates the constitutive relations, the field equations, the boundary conditions at the faces and the edge, and the power flux that identifies a backward mode. The section ends with the dimensionless groups, the reductions the model must reproduce, and the condition for positive definiteness of the elastic–gradient–carrier block.

2.1 Geometry, kinematics and the gradient energy

Consider a transversely isotropic ($6mm$) n-type ZnO plate of thickness d whose faces are the planes $z = \pm d/2$. The plate occupies $x < 0$ and ends at a traction-free edge at $x = 0$. It is unbounded in y, so a single tangential wavenumber k_y is conserved and every field has the form $f(x,z)\,e^{\mathrm{i}k_y y - \mathrm{i}\omega t}$ with all three displacement components retained. The formulation is therefore two-and-a-half-dimensional: the (x,z) cross-section is discretized, y is not. The independent fields, with $\boldsymbol{x} = (x,y,z)$ and the y dependence carried by the factor $\mathrm{e}^{\mathrm{i}k_y y}$ so that the amplitudes depend on (x,z) only, are the displacement $u_i(\boldsymbol{x},t)$, the electrostatic potential $\varphi(\boldsymbol{x},t)$ and the electron-number perturbation $\Delta n(\boldsymbol{x},t)$ measured from the uniform doping n_0. The polar c axis is $x_3 \equiv z$. Latin indices run over $1,2,3$, a comma denotes a spatial derivative and a dot a time derivative. The electric field is $E_i = -\partial_i\varphi$, the electron charge is $-q$ with $q > 0$, and $\widetilde{\mu}$ is the electrochemical potential of an electron per carrier, measured in joules.

The small-strain tensor and its gradient are

$$\varepsilon_{ij} = \tfrac{1}{2}\big(u_{i,j} + u_{j,i}\big), \qquad \eta_{ijk} \equiv \varepsilon_{ij,k} = \tfrac{1}{2}\big(u_{i,jk} + u_{j,ik}\big), \tag{1}$$

so that $\eta_{ijk} = \eta_{jik}$ and η_{ijk} has the dimension of an inverse length. The strain has six independent components and the strain gradient eighteen.

The higher-order elastic energy is taken in the single-length (isotropic-gradient) form

$$w_g = \tfrac{1}{2}\,A_{ijklmn}\,\eta_{ijk}\,\eta_{lmn} = \tfrac{1}{2}\,\ell^2\,c_{ijlm}\,\eta_{ijk}\,\eta_{lmk}, \qquad A_{ijklmn} = \ell^2\,c_{ijlm}\,\delta_{kn}, \tag{2}$$

in which the third and sixth slots of A_{ijklmn} are the gradient directions. This is the contraction that preserves the major symmetry $A_{ijklmn} = A_{lmnijk}$ required of a second derivative of an energy.

The units follow from the derivatives of the energy. Stress $\sigma_{ij} = \partial w/\partial \varepsilon_{ij}$ and stiffness c_{ijkl} are in pascals. The hyperstress $\tau_{ijk} = \partial w/\partial \eta_{ijk}$ is a force per unit length (N/m), the gradient modulus A_{ijklmn} is a force (N), and the flexoelectric constant $f_{ijkl} = \partial D_i/\partial \eta_{jkl}$ is a charge per unit length (C/m). In the carrier sector Δn is a number density ($1/\mathrm{m}^3$), the compressibility $a = k_B T/n_0$ is an energy times a volume, and the coupling $H_{jkl} = \partial \widetilde{\mu}/\partial \eta_{jkl} = \partial \tau_{jkl}/\partial(\Delta n)$ is an energy times a length (J m), the one non-classical unit of the model.

2.2 Constitutive relations and the balance laws

The only physics added to the classical flexoelectric piezoelectric-semiconductor enthalpy is one bilinear term that couples the carrier perturbation to the strain gradient,

$$\Delta \mathcal{H} = \Delta n \, H_{jkl} \, \varepsilon_{jk,l}, \tag{3}$$

where H_{jkl} is a third-order tensor with units of energy times length. The term belongs to a thermodynamic potential, so its two conjugates follow by differentiation and bring in no independent constants. Differentiation with respect to the strain gradient and to the carrier density gives

$$\Delta \tau_{jkl} = \frac{\partial \Delta \mathcal{H}}{\partial \varepsilon_{jk,l}} = H_{jkl} \, \Delta n, \tag{4}$$

$$\Delta \widetilde{\mu} = \frac{\partial \Delta \mathcal{H}}{\partial \Delta n} = H_{jkl} \, \varepsilon_{jk,l}. \tag{5}$$

Only the component $H \equiv H_{zzz}$ is retained in the calculations, so (4) adds $H\Delta n$ to τ_{zzz} and (5) adds $H\eta_{zzz}$ to $\widetilde{\mu}$. Likewise only $f \equiv f_{3zz}$ is retained among the $6mm$-allowed flexoelectric components. The model is therefore a reduced single-component coupling model, not the full $6mm$-invariant flexoelectric–carrier model; the remaining symmetry-allowed components are set to zero and no result reported here tests the sensitivity to that choice.

Relation to the thermodynamic framework. The closest primary work is the thermodynamically consistent theory of Qu, Pan, Zhu and Deng [17], which derives the constitutive equations of a flexoelectric semiconductor from the free-energy imbalance and contains the drift–diffusion current, the flexoelectric and piezoelectric couplings and the higher-order stress in one framework. Its explicit free energy, reproduced in [18], is additive and its carrier energy contains no strain gradient, so in that form the strain gradient reaches the carriers only through the flexoelectric polarization that enters Gauss's law. The bilinear term $\Delta n \, H_{jkl}\varepsilon_{jk,l}$ of (3) is absent from that form but not from the article: its Appendix C.1 adds a chemo-mechanical energy that is linear in the carrier concentration and in the strain gradient, and the part M^n_{IJK} corresponds to H_{jkl} here; the article neglects it in its main text and does not pose an edge-scattering problem. The term is therefore not claimed as new, and what is new is the scattering problem it is used in. The symbol H is overloaded in the literature: in [18] it denotes the sixth-order strain-gradient elasticity tensor, written A_{ijklmn} here, whereas H_{jkl} is a third-order carrier–strain-gradient coefficient (J m) and the two are unrelated. The incremental effect of the term on the edge partition, and the fact that it is not the origin of the backward branch, are quantified in sections 4.5 and 4.6.

The electric enthalpy density of the model is

$$\begin{aligned} w = {} & \tfrac{1}{2}\, \varepsilon_{ij} c_{ijkl} \varepsilon_{kl} + \tfrac{1}{2}\, A_{ijklmn} \eta_{ijk} \eta_{lmn} - \tfrac{1}{2}\, E_i \kappa_{ij} E_j - e_{kij} E_k \varepsilon_{ij} - f_{ijkl} E_i \eta_{jkl} \\ & + \tfrac{1}{2}\, a \, (\Delta n)^2 + H_{jkl} \, \Delta n \, \eta_{jkl} - q \, \Delta n \, \varphi, \end{aligned} \tag{6}$$

with $a = k_B T/n_0$. The terms are, in order, the elastic, strain-gradient, dielectric, piezoelectric and flexoelectric energies of the classical flexoelectric dielectric [11], the carrier compressibility, the added carrier–strain-gradient coupling and the electrostatic energy of the mobile charge. The seventh term, (3), is the one added here, with the structure of the chemo-mechanical term of [17] (Appendix C). Differentiating (6) gives the constitutive relations

$$\sigma_{ij} = \frac{\partial w}{\partial \varepsilon_{ij}} = c_{ijkl}\varepsilon_{kl} - e_{kij}E_k, \tag{7}$$

$$\tau_{ijk} = \frac{\partial w}{\partial \eta_{ijk}} = A_{ijklmn}\eta_{lmn} + H_{ijk}\Delta n - f_{lijk}E_l, \qquad \tau_{ijk} = \tau_{jik}, \tag{8}$$

$$D_i = -\frac{\partial w}{\partial E_i} = \kappa_{ij}E_j + e_{ijk}\varepsilon_{jk} + f_{ijkl}\eta_{jkl}, \tag{9}$$

$$\widetilde{\mu} = \frac{\partial w}{\partial \Delta n} = a\,\Delta n - q\varphi + H_{jkl}\eta_{jkl}. \tag{10}$$

The term-by-term derivatives and the Euler identity for w are given in appendix A. Every term of (10) is an energy: $k_B T\,\Delta n/n_0$ is the chemical part, $-q\varphi$ the electrostatic energy of a particle of charge $-q$, and the H term the energy that the added coupling assigns to a carrier in a strain gradient.

The non-classical entries are reciprocal. The same tensor H_{jkl} appears in Δn in the hyperstress and in η_{jkl} in the electrochemical potential, and the same tensor f_{ijkl} appears in the direct and converse flexoelectric terms of D and τ. This is the Maxwell symmetry of the second derivatives of w, and the discretized operator inherits it: reciprocity at $\pm k_y$ is confirmed numerically to 10^{-13}. A single flexoelectric component, $f \equiv f_{3zz}$, is retained, so that $D_z = e_{zjk}\varepsilon_{jk} + \kappa_{zz}E_z + f\,\eta_{zzz}$ and $\tau_{zzz} = A_{zzz\,lmn}\eta_{lmn} - fE_z$. The surface flexoelectric effect [11] and electric-field-gradient flexoelectric terms are not included.

For the $6mm$ class with the polar axis $x_3 \equiv z$, the tensors reduce to five elastic constants, three piezoelectric constants and two permittivities. In the Voigt ordering $(\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, \gamma_{yz}, \gamma_{xz}, \gamma_{xy})$, with $\gamma_{ij} = 2\varepsilon_{ij}$, the elastic and piezoelectric matrices of ZnO are

$$\mathsf{C} = \begin{pmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{13} & 0 & 0 & 0 \\ c_{13} & c_{13} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{66} \end{pmatrix}, \qquad \begin{aligned} &c_{11} = 210\,\mathrm{GPa}, \quad c_{33} = 221\,\mathrm{GPa}, \\ &c_{13} = 105\,\mathrm{GPa}, \quad c_{12} = 121\,\mathrm{GPa}, \\ &c_{44} = 43\,\mathrm{GPa}, \quad c_{66} = \tfrac{1}{2}(c_{11} - c_{12}), \end{aligned} \tag{11}$$

$$\mathsf{e} = \begin{pmatrix} 0 & 0 & 0 & 0 & e_{15} & 0 \\ 0 & 0 & 0 & e_{15} & 0 & 0 \\ e_{31} & e_{31} & e_{33} & 0 & 0 & 0 \end{pmatrix}, \qquad e_{33} = 1.32\,\mathrm{C/m^2},\ e_{31} = -0.57\,\mathrm{C/m^2},\ e_{15} = -0.48\,\mathrm{C/m^2}, \tag{12}$$

and the permittivity is diagonal, $\kappa_{11} = \kappa_{22}$, $\kappa_{33} = 8.85\times10^{-11}\,\mathrm{F/m}$. The component relations used are

$$\begin{aligned} \sigma_{xx} &= c_{11}\varepsilon_{xx} + c_{12}\varepsilon_{yy} + c_{13}\varepsilon_{zz} - e_{31}E_z, \\ \sigma_{yy} &= c_{12}\varepsilon_{xx} + c_{11}\varepsilon_{yy} + c_{13}\varepsilon_{zz} - e_{31}E_z, \\ \sigma_{zz} &= c_{13}(\varepsilon_{xx} + \varepsilon_{yy}) + c_{33}\varepsilon_{zz} - e_{33}E_z, \\ \sigma_{yz} &= 2c_{44}\varepsilon_{yz} - e_{15}E_y, \qquad \sigma_{xz} = 2c_{44}\varepsilon_{xz} - e_{15}E_x, \qquad \sigma_{xy} = 2c_{66}\varepsilon_{xy}, \end{aligned} \tag{13}$$

$$\tau_{zzz} = \ell^2\Big[c_{13}\big(\eta_{xxz} + \eta_{yyz}\big) + c_{33}\,\eta_{zzz}\Big] - f\,E_z + H\,\Delta n, \tag{14}$$

$$
\begin{aligned}
D_x &= \kappa_{11} E_x + 2 e_{15} \varepsilon_{xz}, \qquad D_y = \kappa_{11} E_y + 2 e_{15} \varepsilon_{yz}, \\
D_z &= \kappa_{33} E_z + e_{31} (\varepsilon_{xx} + \varepsilon_{yy}) + e_{33} \varepsilon_{zz} + f\, \eta_{zzz}.
\end{aligned}
\tag{15}
$$

The field equations are of fourth order in the mechanical sector and of second order in the electrical and carrier sectors, which is why the plate supports four wave families per branch.

The field equations are the Euler–Lagrange equations of the enthalpy. The variations $\delta\varepsilon_{ij}$ and $\delta\eta_{ijk} = \tfrac{1}{2}(\delta u_{i,jk} + \delta u_{j,ik})$ follow from (1), and because $\tau_{ijk} = \tau_{jik}$ one has $\tau_{ijk}\,\delta\eta_{ijk} = \tau_{ijk}\,\delta u_{i,jk}$. One integration by parts on the stress and two on the hyperstress move all derivatives onto the fields, and the surface terms are the traction and the double traction,

$$
\int_V \tau_{ijk}\,\delta\eta_{ijk}\,\mathrm{d}V = \int_V \partial_k\partial_j\tau_{ijk}\,\delta u_i\,\mathrm{d}V - \oint_{\partial V} n_j\partial_k\tau_{ijk}\,\delta u_i\,\mathrm{d}S + \oint_{\partial V} n_k\tau_{ijk}\,\delta u_{i,j}\,\mathrm{d}S. \tag{16}
$$

With the inertial term, the mechanical balance is

$$
\rho\,\ddot{u}_j = \partial_i\sigma_{ij} - \partial_i\partial_k\tau_{jik}, \tag{17}
$$

where the double divergence of the hyperstress enters with a minus sign because the gradient energy acts against the elastic restoring force. With the effective stress

$$
\Sigma_{ij} = \sigma_{ij} - \partial_k\tau_{ijk}, \tag{18}
$$

the balance reads $\rho\ddot{u}_j = \partial_i\Sigma_{ij}$ and the traction on a surface is $t_j = n_i\Sigma_{ij}$. The expanded out-of-plane component of (17) and of Gauss's law are given in appendix A.

Variation with respect to φ gives Gauss's law,

$$
\partial_i D_i = -q\,\Delta n. \tag{19}
$$

The carrier equation is not a variational equation of the enthalpy: the derivative with respect to Δn only defines $\widetilde{\mu}$ in (10), and the continuity statement follows from conservation of carrier number with the drift–diffusion flux,

$$
\partial_t(\Delta n) + \partial_i J_i = 0, \qquad J_i = -\frac{D\,n_0}{k_B T}\,\partial_i\widetilde{\mu}, \tag{20}
$$

where J_i is the electron number flux and $D = \mu_e k_B T/q$ the diffusivity (Einstein relation). With $\partial_t \to -\mathrm{i}\omega$ this gives $\Delta n = \mathrm{i} D n_0 \nabla^2\widetilde{\mu}/(\omega k_B T)$, and the factor i reappears as $+\mathrm{i}\alpha_\mathrm{d}/\omega$ in the weak form (38); reversing its sign would turn the carrier sector from a sink of energy into a source. The charge-current density is

$$
j_i = -qJ_i = \frac{qDn_0}{k_B T}\,\partial_i\widetilde{\mu} = qD\,\partial_i\Delta n - q\mu_e n_0\,\partial_i\varphi + \frac{qDn_0}{k_B T}\,H_{jkl}\,\partial_i\eta_{jkl}, \tag{21}
$$

whose three terms are the diffusion current, the drift current and the carrier–strain-gradient contribution. The last term is the transport signature of (3): a strain-gradient field drives a current even at zero applied field.

2.3 The electric model, the boundary conditions and the energy flux

Equations (19)–(20) are electrostatic: no magnetic induction, displacement current or electromagnetic radiation mode is retained. This is justified by the ratios that control it. At $f = 16.68\,\mathrm{GHz}$ and with $\kappa_{33}/\varepsilon_0 = 10$ the electromagnetic wavenumber is $k_\mathrm{EM} =$

$1.106 \times 10^3/\mathrm{m}$, so $k_{\mathrm{EM}}d = 2.2 \times 10^{-4}$ for $d = 200\,\mathrm{nm}$ and the retardation correction is of order $(k_{\mathrm{EM}}d)^2 = 4.9 \times 10^{-8}$. The phase speed gives the same result: $v_p/c = 9.4 \times 10^{-5}$, so the magnetic energy is smaller than the electric energy by $(v_p/c)^2 \approx 8.8 \times 10^{-9}$.

The carriers are different. With $\sigma_e = q\mu_e n_0 = 5.6 \times 10^{-3}\,\mathrm{S/m}$ the dielectric relaxation time is $\tau_d = \kappa_{33}/\sigma_e = 1.58 \times 10^{-8}\,\mathrm{s}$ against a wave period of $6.0 \times 10^{-11}\,\mathrm{s}$, so $\omega\tau_d = 1.65 \times 10^3$ and the carriers cannot follow the field quasistatically. The model therefore keeps the full continuity equation (20); what is neglected is electromagnetic retardation, not charge dynamics. The Debye length is $L_D = \sqrt{\kappa_{33}k_BT/(q^2n_0)} = 2.68\,\mu\mathrm{m}$, so $d/L_D = 0.075$ and the carrier perturbation is not confined to a surface layer.

The boundary conditions follow from the surface terms of the variational principle.

Plate faces $z = \pm d/2$. The faces are free surfaces, so the generalized traction of (26) (with face normal z) and the double traction vanish; the electrons cannot leave the crystal; and the electrical condition is grounded or open circuit,

$$\bar{\sigma}_{iz} = 0, \qquad \tau_{izz} = 0, \qquad z = \pm d/2, \tag{22}$$

$$J_z = 0, \qquad z = \pm d/2, \tag{23}$$

$$\varphi = 0 \quad \text{(grounded)}, \qquad D_z = 0 \quad \text{(open circuit)}, \qquad z = \pm d/2. \tag{24}$$

The grounded condition models perfectly conducting electrodes shorted to each other; it is essential and is imposed on the basis functions. The open-circuit condition models a bare surface with no free charge; it is natural and is imposed weakly, with no potential mode removed. A potential that is constant across the thickness is a gauge only in the sector $k_x = k_y = 0$, which is not used here; for $(k_x, k_y) \neq (0, 0)$ it gives physical in-plane fields. The open-circuit results are checked against the independent C^1 discretization, which tests the discretization and not the physics.

Reflecting edge $x = 0$. The edge is a free surface in vacuum and no datum is prescribed on it, so all four conditions are natural, each acting on the test functions of the field it is conjugate to,

$$\bar{\sigma}_{ix} = 0, \qquad \tau_{ixx} = 0, \qquad D_x = 0, \qquad J_x = 0, \qquad x = 0, \tag{25}$$

where

$$\bar{\sigma}_i(\boldsymbol{n}) = n_k \underbrace{\big(\sigma_{ik} - \partial_l \tau_{ikl}\big)}_{\Sigma_{ik}} - \partial_j^s\big(n_k \tau_{ijk}\big), \tag{26}$$

is the generalized traction on a surface of unit normal $\boldsymbol{n}$, with ∂_j^s the derivative tangential to that surface: the effective stress minus the surface divergence of the hyperstress. For the edge $\boldsymbol{n} = \boldsymbol{e}_x$ it is written $\bar{\sigma}_{ix}$, and for the faces $\boldsymbol{n} = \pm\boldsymbol{e}_z$ it gives $\pm\bar{\sigma}_{iz}$, with $j = x, y$ tangential there. The free index is the first index of the hyperstress and its gradient index is contracted with the normal, so the edge term is $\partial_j^s \tau_{ijx}$ and not $\partial_j^s \tau_{ixj}$; a wrong index changes the result. With $n = e_x$ and a tangential wavenumber k_y, $\bar{\sigma}_{jx} = \Sigma_{jx} - \mathrm{i}k_y\tau_{jyx} - \partial_z\tau_{jzx}$. The second entry of (25) is the double traction, conjugate to the normal derivative of the displacement on the edge. In this model $\tau_{ijx} = \ell^2 c_{ijkl}\eta_{klx}$, so the double traction is present if and only if $\ell \neq 0$; no result reported here takes the limit $\ell = 0$ with $f \neq 0$. The potential is free on the edge, so $D_x = 0$ is its natural condition. The surviving terms are set out case by case in appendix A.

Far field $x \to -\infty$. The solution is the prescribed incident guided mode plus outgoing branches; no other wave arrives from $x = -\infty$.

The corner. Where the edge meets a face, the surface divergence in (26) has a boundary, and the integration by parts that produced the traction also produces a line force along the corner. With ν the co-normal of the edge boundary inside the face, its density per unit length is

$$f_i = \nu_j \, n_k \, \tau_{jik}, \tag{27}$$

summed over the two free faces that meet along the corner. For this geometry the contributions are the mixed components τ_{i31} and τ_{i13}, so $f_i = \tau_{i31} + \tau_{i13}$. The relative sign and the index placement are fixed, together with (26) and the double traction $m_i = \tau_{i11}$, by the identity

$$\Big\langle T_u - t \Big\rangle = f_+ \, b_0(+\tfrac{1}{2}) - f_- \, b_0(-\tfrac{1}{2}), \qquad \Big\langle R - m \Big\rangle = 0, \tag{28}$$

between the derived fields and the discrete flux rows T_u, R of the canonical state, with $\langle\cdot\rangle$ the projection on the displacement test basis b_0. The identity holds to 1.4×10^{-13} relative for the incident guided mode at all degrees from 10 to 28, and it is violated at the level of 1.7×10^{1} if the corner terms are dropped, at 5.7×10^{-1} if only τ_{i31} is kept, and at 1.1×10^{0} if the two are subtracted instead of added, so the corner line force is an essential part of the natural data. It is the one natural datum that a polynomial Galerkin basis cannot annihilate pointwise; its magnitude and decay are quantified in the conclusions.

The power flowing in the x direction, per unit width and integrated over the thickness, is

$$\mathcal{P}_x = \tfrac{1}{2} \operatorname{Re} \int_{-d/2}^{d/2} \Big[-\Sigma_{xi} \dot{u}_i^* - \tau_{jkx} \dot{\varepsilon}_{jk}^* + \varphi \dot{D}_x^* + \widetilde{\mu} \, J_x^* \Big] \mathrm{d}z - \tfrac{1}{2} \operatorname{Re} \Big[\tau_{ixz} \dot{u}_i^* \Big]_{z=-d/2}^{z=d/2}, \tag{29}$$

where Σ_{xi} is the effective stress (18) and J_x is the x component of the electron-number flux of (20). The four integrands are the elastic, hyperelastic, dielectric and chemical contributions, and the last term is the line work of the double stress on the two faces. It does not vanish for a guided mode and it is needed for the flux to balance the loss: without it the single-mode balance $|2 \operatorname{Im} k_x \, \mathcal{P}_x| = P_{\mathrm{diss}}$ is violated at the tens-of-percent level, and with it the balance closes to 3×10^{-9} or better. The flux $\mathcal{P}_x$, and not the magnitude of the wavevector, decides whether a mode is backward: a mode is backward when $\mathcal{P}_x < 0$ while it is outgoing at the edge.

For a scattering solution the identity

$$\underbrace{P_{\mathrm{inc}}}_{\text{incident}} + \underbrace{\sum_j P_j}_{\text{self-powers}} + \underbrace{P_{\mathrm{cross}}}_{\text{interference}} = \underbrace{P_{\mathrm{edge}} + P_{\mathrm{diss}}}_{\text{edge flux plus loss}} \tag{30}$$

holds exactly, with P_j the signed self-power of outgoing mode j, P_{cross} the bilinear interference between modes that are not orthogonal in the power metric, and

$$P_{\mathrm{diss}} = \tfrac{1}{2} \alpha_{\mathrm{d}} \int_{-L}^{0} \!\! \int \Big[|\partial_x \widetilde{\mu}|^2 + k_y^2 |\widetilde{\mu}|^2 + |\partial_z \widetilde{\mu}|^2 \Big] \mathrm{d}z \, \mathrm{d}x, \qquad \alpha_{\mathrm{d}} = \frac{D}{d \, c_s},$$

written in scaled variables, with $\widetilde{\mu}$ the total electrochemical-potential field divided by $k_B T$. The integrand is nonnegative, so $P_{\mathrm{diss}} \geq 0$, and α_{d} is the ratio of the diffusion rate to the elastic rate, a property of the material and thickness and not of a mode (the spatial attenuation of a mode is $\alpha = 2|\operatorname{Im} k_x|$). Only the propagating members of $\{P_j\}$ transport energy to infinity; the evanescent members are reactive at the plane and P_{cross} is an interference correction. The residual of (30) is evaluated for each reported solution; its values at the operating point for the five coupling levels are plotted in fig. 5(c).

2.4 Plane-wave reduction and the characteristic polynomial

The plate is invariant along y and in time, so a single tangential wavenumber and frequency are conserved and every field takes the form $f = \widehat{f}(z)\,\mathrm{e}^{\mathrm{i}(k_x x + k_y y - \omega t)}$, for which

$$\partial_t \to -\mathrm{i}\omega, \qquad \partial_x \to \mathrm{i}k_x, \qquad \partial_y \to \mathrm{i}k_y, \qquad \partial_z \to \frac{\mathrm{d}}{\mathrm{d}z} \equiv {}'. \tag{31}$$

The strain, strain-gradient and field components follow by applying (31) to (1); for example $\widehat{\varepsilon}_{xx} = \mathrm{i}k_x\widehat{u}_1$, $\widehat{\eta}_{xxx} = -k_x^2\widehat{u}_1$, $\widehat{\eta}_{zzz} = \widehat{u}_3''$ and $\widehat{E}_x = -\mathrm{i}k_x\widehat{\varphi}$. The continuity equation (20) becomes algebraic, so the carrier amplitude is carried by the same amplitude vector as the displacements and has no independent time evolution.

The dependence on k_x is fixed by derivative counting. The stress σ is linear in ε and E, each containing one power of k_x, and enters the mechanical balance through one further derivative, so that term is at most quadratic in k_x. The hyperstress τ is linear in η, which contains two powers, and enters through two derivatives, so that term is at most quartic. Counted on the assembled operator, each side of the weak form carries at most two powers (one from ε or E, two from η, none from Δn or $\widetilde{\mu}$), so no power above four arises, and four arises only from the η–η gradient energy. The retained couplings add nothing at that order: $H\,\Delta n\,\eta_{zzz}$ pairs a zero-power factor with $\eta_{zzz} = \widehat{u}_3''$, which contains no k_x, and $f_{3zz}E_z\eta_{zzz}$ involves only z derivatives. For a vector $\boldsymbol{q}$ of thickness degrees of freedom the reduced problem is therefore a polynomial of degree four in k_x,

$$\Big[\mathsf{A}_{00} + \mathrm{i}k_x\big(\mathsf{A}_{01} - \mathsf{A}_{10}\big) + k_x^2\mathsf{A}_{11} + \mathrm{i}k_x^3\big(\mathsf{A}_{12} - \mathsf{A}_{21}\big) + k_x^4\,\mathsf{A}_{22}\Big]\boldsymbol{q} = \boldsymbol{0}, \tag{32}$$

where A_{rs} collects the terms containing ∂_x^r on the left test function and ∂_x^s on the right one (their construction is given with the discretization), and the k_x^2 coefficient reduces to A_{11} because $\mathsf{A}_{02}+\mathsf{A}_{20}$ does not arise from this enthalpy. The odd coefficients are Hermitian, being i times antisymmetric differences; the even ones carry the non-Hermitian carrier transport form. The degree is exactly four only for $\ell \neq 0$: for the retained f_{3zz} and H_{zzz}, A_{22} and A_{12} come from the strain-gradient energy alone, so for $\ell = 0$ (with f and H removed) $\mathsf{A}_{22} = \mathsf{A}_{12} = 0$ and (32) collapses to the quadratic problem of the classical piezoelectric-semiconductor plate. The fourth-order structure is thus a property of the strain-gradient energy, not of the carrier physics.

Read by sectors and not by powers of k_x, the same operator has a 3×3 block structure. With $\mathsf{B}^{,l}$ the strain operator differentiated once more along $l = x, y, z$ and unknowns ordered as $(\widehat{\boldsymbol{u}}, \widehat{\boldsymbol{\varphi}}, \widetilde{\boldsymbol{\mu}})$,

$$\mathsf{A}(k_x,\omega) = \begin{pmatrix} \mathsf{A}_{uu} & \mathsf{A}_{u\varphi} & \mathsf{A}_{u\mu} \\ \mathsf{A}_{\varphi u} & \mathsf{A}_{\varphi\varphi} & \mathsf{A}_{\varphi\mu} \\ \mathsf{A}_{\mu u} & \mathsf{A}_{\mu\varphi} & \mathsf{A}_{\mu\mu} \end{pmatrix}, \qquad \begin{aligned} \mathsf{A}_{uu} &= \langle \mathsf{B}, \mathsf{C}\mathsf{B}\rangle + \lambda^2 \sum_l \langle \mathsf{B}^{,l}, \mathsf{C}\mathsf{B}^{,l}\rangle - \omega^2\langle \mathsf{U}, \mathsf{U}\rangle, \\ \mathsf{A}_{u\varphi} &= \langle \mathsf{B}, \mathsf{e}^{\mathsf{T}}\mathsf{G}\rangle + f\langle \partial_\zeta \mathsf{B}^{(2)}, \mathsf{G}^{(2)}\rangle, \\ \mathsf{A}_{\varphi\varphi} &= -\langle \mathsf{G}, \boldsymbol{\kappa}\mathsf{G}\rangle, \qquad \mathsf{A}_{\varphi\mu} = -Q\langle \mathsf{G}^{(0)}, \mathsf{B}\rangle, \\ \mathsf{A}_{\mu\mu} &= -\langle \mathsf{W}, \mathsf{W}\rangle - \frac{\mathrm{i}\alpha_{\mathrm{d}}}{\omega}\langle \mathsf{M}\mathsf{G}, \mathsf{M}\mathsf{G}\rangle, \end{aligned} \tag{33}$$

where the carrier form $-\langle \mathsf{W}, \mathsf{W}\rangle$ of (39), with $\mathsf{W} = (-h\,b_j''$ in the $\widehat{u}_3$ slots, $Q\psi_j,\ b_j)$ and the prime the derivative with respect to z/d, also contributes $-h^2\langle b'', b''\rangle$ to A_{uu}, $+hQ\langle b'', \psi\rangle$ to $\mathsf{A}_{u\varphi}$, $+h\langle b'', b\rangle$ to $\mathsf{A}_{u\mu}$ and $-Q^2\langle \psi, \psi\rangle$ to $\mathsf{A}_{\varphi\varphi}$. The remaining blocks are the Hermitian transposes of those written, being the same derivatives of the same enthalpy taken in the other order. The symmetric part of (33) is the Hessian of the discretized enthalpy; the carrier transport form is the only antisymmetric contribution and the entire source of loss.

Table 1: Dimensionless groups and their values at the operating point; $T = 301.7\,\mathrm{K}$ follows from the Einstein relation; h is given at the largest coupling examined, $|H| = 2.0 \times 10^{-23}\,\mathrm{J\,m}$.

Group	Definition	Value	Meaning
Q	d/L_D	7.46×10^{-2}	thickness over Debye length
α_d	$D/(d\,c_s)$	8.28×10^{-1}	diffusion rate over elastic rate
λ	ℓ/d	2.50×10^{-1}	gradient length over thickness
f	$f_{3zz}/(d\sqrt{c_{44}\kappa_{33}})$	2.56×10^{-3}	flexoelectric coupling
h	$H/(d\sqrt{a\,c_{44}})$	$\pm 1.06 \times 10^{-2}$	carrier–gradient coupling (at $\lvert H\rvert_\mathrm{max}$)

2.5 Non-dimensionalization, limits and admissibility

The calculation uses lengths scaled by d, stresses by c_{44}, time by d/c_s with $c_s = \sqrt{c_{44}/\rho}$, carrier densities by n_0 and energies per carrier by k_BT. The carrier amplitude of (20) is measured so that the compressibility term of (6) is $\frac{1}{2}\int \widehat{N}^2$ in the scaled units used by (39), which is what fixes the coefficient of $\widehat{N}^2$ there. Two derived scales appear repeatedly,

$$a = \frac{k_BT}{n_0}, \qquad L_D = \sqrt{\frac{\kappa_{33}k_BT}{q^2 n_0}},$$

and the model is governed by the five dimensionless numbers of table 1. Each is an exact ratio of two physical lengths, times or energies, and each follows directly from the parameters listed below.

Parameters and their provenance. The classical constants are literature values ($c_{11} = 210$, $c_{33} = 221$, $c_{44} = 43\,\mathrm{GPa}$; $e_{33} = 1.32$, $e_{31} = -0.57$, $e_{15} = -0.48\,\mathrm{C/m^2}$; $\kappa_{33} = 8.85 \times 10^{-11}\,\mathrm{F/m}$, from Table 1 of [6]). The transport parameters are representative values for n-type ZnO: $\mu_e = 0.0175\,\mathrm{m^2/(V\,s)}$ and $D = 4.55 \times 10^{-4}\,\mathrm{m^2/s}$, equal to $\mu_e k_BT/q$ at $T = 301.7\,\mathrm{K}$ and within the 120–205 $\mathrm{cm^2/(V\,s)}$ compiled for nominally undoped bulk and film ZnO in Table VII of [2]; the density $n_0 = 1$–$3 \times 10^{18}\,1/\mathrm{m^3}$ is a modeling scenario below the 8×10^{19}–$5 \times 10^{23}\,1/\mathrm{m^3}$ of that table, and $d = 200\,\mathrm{nm}$ is a reference thickness (sweep 120–320 nm). The strain-gradient length $\ell = 50\,\mathrm{nm}$, the flexoelectric coefficient $f_{3zz} = 1\,\mathrm{nC/m}$ (in the nC/m range reported for ZnO [18, 44]; it contributes $-0.01\,\%$ to R_B) and the coupling H_{333} are *assumed*; the same form $g = \ell^2 c$ is used in [45] with 20 nm for silicon. For H_{jkl} no published value is available: the numbers used, $|H| \le 2.0 \times 10^{-23}\,\mathrm{J\,m} = 0.022\,H_\mathrm{bound}$, are fixed by the positivity bound (34), which leaves a margin of a factor of forty-five in H.

The groups follow from the listed parameters by algebra, so their verification is a comparison of numbers and not a fit. The elastic speed is $c_s = \sqrt{c_{44}/\rho} = 2.7466\,\mathrm{km/s}$, giving the frequency unit $c_s/(2\pi d) = 2.1857\,\mathrm{GHz}$ at $d = 200\,\mathrm{nm}$; the operating point $\omega = 7.6334\,c_s/d$ is then $f = 16.684\,\mathrm{GHz}$, the value quoted in the abstract. With $a = k_BT/n_0 = 2.0828 \times 10^{-39}\,\mathrm{Jm^3}$ (the temperature $T = 301.72\,\mathrm{K}$ is implied by the Einstein relation for the diffusivity–mobility pair, not an independent input), the Debye length is $L_D = 2.6797\,\mu\mathrm{m}$ and $Q = d/L_D = 0.074635$. The plate is thus thinner than $L_D/d = 13.40$ times a screening length, the carrier perturbation is not confined to a surface layer, and the screening enters (39) only through the small coefficient $Q^2 = 5.57 \times 10^{-3}$. The transport group is $\alpha_\mathrm{d} = D/(d\,c_s) = 0.8283$, of order unity: the diffusion time across the thickness, $d^2/D = 8.8 \times 10^{-11}\,\mathrm{s}$, is comparable to the elastic transit time $2d/c_s = 1.5 \times 10^{-10}\,\mathrm{s}$, so the carrier and elastic sectors are not separated by a small parameter and the carrier cannot be condensed away. Finally $\lambda = \ell/d = 0.25$ and $h = H/H_\mathrm{ref}$ with $H_\mathrm{ref} = d\sqrt{a\,c_{44}} = 1.8927 \times 10^{-21}\,\mathrm{J\,m}$,

which has the units of H required by (6); the tabulated extreme $|H| = 2.0 \times 10^{-23}\,\mathrm{J\,m}$ is $h = \pm 1.06 \times 10^{-2}$, a one-percent perturbation of the carrier sector.

The model of (6) contains four sectors that can be switched off independently, and the limits they generate are the ones against which the calculation is read.

Classical piezoelectric plate. Setting $\ell = f = H = 0$ and removing the carrier sector ($\Delta n \equiv 0$) leaves the classical piezoelectric formulation, with a polynomial in k_x of degree two instead of four.

Strain-gradient dielectric. With $H = 0$ and no carriers the enthalpy is the flexoelectric dielectric enthalpy of [11] completed by the gradient energy (2). In this limit the fundamental shear-horizontal branch has the closed form $\Omega^2 = (c_{66}/c_{44})K^2(1 + \lambda^2 K^2)$ in the scaled units of this subsection, which is used as one of the checks.

Piezoelectric semiconductor without the gradient sector. Setting $\ell = f = H = 0$ and retaining the carriers leaves a drift–diffusion plate whose sagittal spectrum is examined in section 4.5: at the frequencies of the controls it contains no outgoing mode with a positive axial wavenumber and a negative axial power apart from the specular reflection. The plate nevertheless has a backward interval of its own; the gradient length places the fold of that interval at the reported frequency.

Uncoupled carriers. With $H = 0$ and the strain-gradient sector retained, the carrier dynamics still enter through Gauss's law and through the continuity equation (20), so the plate remains a piezoelectric semiconductor; only the direct carrier–gradient enthalpy of (3) is removed. This is the limit in which the controls are computed, and it is the limit that shows the partition to be a property of the branch and not of the added term.

The enthalpy (6) is a quadratic form in $(\varepsilon, \eta, E, \Delta n)$. It is a saddle in the electric field, because the term $-\frac{1}{2}E_i\kappa_{ij}E_j$ is negative, so the full form is not positive definite and none of the statements below concerns it. The block in $(\varepsilon, \eta, \Delta n)$ at fixed E is positive definite only if the carrier and strain-gradient parts are not over-coupled. Eliminating Δn blockwise gives the Schur condition

$$s \equiv \frac{H^2\,(c^{-1})_{33}}{\ell^2\,a} < 1, \tag{34}$$

where $(c^{-1})_{33}$ is the 33 entry of the inverse of the full elastic matrix, i.e. the reduced modulus $c^* = c_{33} - 2c_{13}^2/(c_{11} + c_{12}) = 154.4\,\mathrm{GPa}$ and not c_{33} itself. The bound is attained at $H_{\mathrm{bound}} = \ell\sqrt{a\,c^*} = 8.97 \times 10^{-22}\,\mathrm{J\,m}$, which is the value at which the coupled stiffness of the plate ceases to be positive for every strain-gradient field. Every calculation reported below is confined to (34). All reported solutions have $|H| \le 2.0 \times 10^{-23}\,\mathrm{J\,m} = 0.022\,H_{\mathrm{bound}}$, i.e. $s \le 5 \times 10^{-4}$: the $(\varepsilon, \eta, \Delta n)$ block is positive definite with a margin of more than two thousand, and no reported result lies near a constitutive instability.

The figures normalize the coupling by $H^* = H_{\mathrm{bound}}/\sqrt{5} = 4.01 \times 10^{-22}\,\mathrm{J\,m}$, which is the value at which $s = 0.2$; the normalization is a plotting convention with no physical content beyond fixing the swept interval $|H/H^*| \le 0.05$, that is $|H| \le 2.0 \times 10^{-23}\,\mathrm{J\,m}$, the tabulated extreme. The separate symbol $H_{\mathrm{ref}} = d\sqrt{a\,c_{44}} = 1.89 \times 10^{-21}\,\mathrm{J\,m}$ used in table 1 is the thickness-based reference of (6); it is not the plotting normalization, and the two differ. No strain-gradient elasticity measurement for ZnO fixes any of ℓ, f or H, so the reported intervals are modeling choices and have no experimental uncertainty attached; the numerical uncertainty of a reported R_B is its refinement spread, below 3×10^{-5} in relative terms.

3 Solution of the problem

The edge problem is solved at fixed frequency. The outgoing spectrum is the set of eigenvalues of the spatial problem, and the partition among those modes follows from the free-edge

conditions. The thickness direction is discretized first, then the modes are obtained, classified and matched at the edge, and independent checks close the section.

3.1 Thickness discretization and the assembly of the blocks

The thickness dependence is represented by a spectral Galerkin expansion. Because the enthalpy contains the strain gradient, the weak form contains second derivatives through the thickness and the trial space must be at least C^1 in z. Legendre polynomials are used for all fields, in the scaled variable $\zeta = 2z/d \in [-1, 1]$,

$$U_i(x,\zeta) = \sum_{j=0}^{n} U_{ij}(x)\, b_j(\zeta), \qquad N(x,\zeta) = \sum_{j=0}^{n} N_j(x)\, b_j(\zeta), \qquad b_j = \sqrt{2j+1}\, P_j, \tag{35}$$

with P_j the Legendre polynomial of degree j, b_j the orthonormal scaled form of (50), n the polynomial degree and $N = n + 1$ the number of basis functions per field. The two electrical cases use different potential spaces, which is how the essential and natural conditions are realized:

- grounded, (24): the potential basis is formed from the combination $\psi_j = b_j - \sqrt{(2j+1)/(2j+5)}\, b_{j+2}$, which vanishes at $\zeta = \pm 1$ for all j because $b_j(\pm 1) = (\pm 1)^j \sqrt{2j+1}$, so $\varphi = 0$ holds on both faces to machine precision and the space has $N - 2$ members;
- open circuit, the second condition of (24): all N basis functions are kept and unconstrained, so $D_z = 0$ is imposed weakly through the surface term of the weak form.

The mechanical conditions (22) and the blocking condition (23) are natural and appear as the corresponding boundary rows of the discretized operator. The basis is integrated with $\max(64, 2n + 10)$ Gauss–Legendre points, which resolves the mass and stiffness matrices to machine precision for all degrees used here.

The expansion (35) is inserted in the weak equations of the reduced problem. There are three of them, and they are kept apart here because they are three different statements: the mechanical balance, Gauss's law and the carrier continuity. Let $(\boldsymbol{v}, \psi, \Phi)$ be test fields of the same spaces as $(\boldsymbol{u}, \varphi, \widetilde{\mu})$.

(A) Mechanical balance.

$$\int_{-1}^{1} \Big[\widehat{\sigma}_{ij}\, \widehat{\varepsilon}^{*}_{ij}(\boldsymbol{v}) + \widehat{\tau}_{ijk}\, \widehat{\eta}^{*}_{ijk}(\boldsymbol{v}) \Big] \mathrm{d}\zeta = \omega^2 \int_{-1}^{1} \rho\, \widehat{u}_i\, v^{*}_i \, \mathrm{d}\zeta, \tag{36}$$

with $\widehat{\sigma}$ and $\widehat{\tau}$ the trial fields of (7)–(8). The boundary terms produced by integrating $\partial_j \widehat{\sigma}_{ij}$ once and $\partial_j \partial_k \widehat{\tau}_{ijk}$ twice vanish on the free, double-traction-free faces (22); they are the traction and the double traction.

(B) Gauss's law.

$$-\int_{-1}^{1} \widehat{D}_i\, \widehat{E}^{*}_i(\psi)\, \mathrm{d}\zeta - Q \int_{-1}^{1} \widehat{N}\, \psi^{*}\, \mathrm{d}\zeta = 0, \tag{37}$$

with the trial $\widehat{D}$ from (9), $\widehat{N}$ the dimensionless carrier amplitude and $Q = d/L_D$ the dimensionless charge of table 1 that the scaling puts in front of $-q\Delta n$. The second integral is the mobile charge of (19); without it (37) would be the statement for a dielectric rather than Gauss's law. The grounded condition is essential and is built into the potential space; the open-circuit condition $D_z = 0$ is the natural boundary term of (37) and is imposed through it.

(C) Carrier continuity.

$$\int_{-1}^{1}\Big[\widehat{N}\,\Phi^{*}+\frac{\mathrm{i}\alpha_{\mathrm{d}}}{\omega}\,\partial_i\widetilde{\mu}\,\partial_i\Phi^{*}\Big]\mathrm{d}\zeta=0, \tag{38}$$

where $\widehat{N}=\widetilde{\mu}+Q\widehat{\varphi}-h\,\widehat{u}_3''$ is the carrier amplitude that (20) delivers (the prime is the derivative with respect to z/d); the last term comes from $H\eta_{zzz}$ in (10) and vanishes when $H=0$. Both signs are fixed by the time convention $\partial_t\to-\mathrm{i}\omega$: multiplying $-\mathrm{i}\omega\widehat{N}-\alpha_{\mathrm{d}}\,\partial_i\partial_i\widetilde{\mu}=0$ by i/ω and using $\mathrm{i}(-\mathrm{i})=1$ gives $\widehat{N}-(\mathrm{i}\alpha_{\mathrm{d}}/\omega)\partial_i\partial_i\widetilde{\mu}=0$, and one integration by parts moves the gradient term to the plus sign displayed above; the boundary term $(\mathrm{i}\alpha_{\mathrm{d}}/\omega)\partial_z\widetilde{\mu}\,\Phi^{*}$ which vanishes with the blocking condition $J_z=0$; this condition is therefore natural for (38). It is a transport equation, not a statement about the enthalpy: the enthalpy reaches it only through $\widetilde{\mu}$, which is why the coupling also appears in (36), through $\widehat{\tau}$.

(D) The assembled system. Because the three equations are coupled they are discretized in one matrix. The unknown vector is $\boldsymbol{q}=(\widehat{\boldsymbol{u}},\widehat{\boldsymbol{\varphi}},\widetilde{\boldsymbol{\mu}})$ of dimension $3n+n_{\varphi}+n$, and the matrix is stored as nine blocks A_{rs}, $r,s\in\{0,1,2\}$, indexed by the power of k_x carried by the test factor (r) and by the trial factor (s). With B_r, G_r and MG_r the strain, electric-field and electrochemical-gradient operators carrying ∂_x^r, with W the row vector that returns the carrier amplitude $\widehat{N}=\widetilde{\mu}+Q\widehat{\varphi}-h\,\widehat{u}_3''$ from the unknowns (its entries are b_j in the $\widetilde{\mu}$ slots, $Q\psi_j$ in the potential slots and $-h\,b_j''$ in the $\widehat{u}_3$ slots), so that $\langle\mathsf{W},\mathsf{W}\rangle$ carries the compressibility $a(\Delta n)^2$, the charge Q and the coupling H, and with U the displacement basis, each block is the bilinear form

$$\begin{aligned}\mathsf{A}_{rs}=&\Big\langle\mathsf{B}_r,\mathsf{C}\,\mathsf{B}_s\Big\rangle+\lambda^2\sum_{l}\Big\langle\mathsf{B}_r^{,l},\mathsf{C}\,\mathsf{B}_s^{,l}\Big\rangle+\Big\langle\mathsf{G}_r,\mathsf{e}\,\mathsf{B}_s\Big\rangle+\Big\langle\mathsf{B}_r,\mathsf{e}^{\mathsf{T}}\mathsf{G}_s\Big\rangle-\Big\langle\mathsf{G}_r,\boldsymbol{\kappa}\,\mathsf{G}_s\Big\rangle\\&+f\Big[\Big\langle\mathsf{G}_r^{(2)},\partial_\zeta\mathsf{B}_s^{(2)}\Big\rangle+\Big\langle\partial_\zeta\mathsf{B}_r^{(2)},\mathsf{G}_s^{(2)}\Big\rangle\Big]-\frac{\mathrm{i}\alpha_{\mathrm{d}}}{\omega}\Big\langle\mathsf{MG}_r,\mathsf{MG}_s\Big\rangle-\Big\langle\mathsf{W}_r,\mathsf{W}_s\Big\rangle-\omega^2\Big\langle\mathsf{U}_r,\mathsf{U}\Big\rangle\delta_{r0}\delta_{s0},\end{aligned} \tag{39}$$

in the inner product $\langle\boldsymbol{a},\boldsymbol{b}\rangle=\int_{-1/2}^{1/2}\boldsymbol{a}^{*\mathsf{T}}\boldsymbol{b}\,\mathrm{d}s$ over the half-interval coordinate (49), with the complex conjugate of the test factor, and where $\mathsf{B}^{,l}$ is the strain operator differentiated once more along $l=x,y,z$ ($\mathrm{i}k_x$ through the index r,s, $\mathrm{i}k_y$, and ∂_ζ), and with the eight terms, in order, the elastic, strain-gradient, piezoelectric (two cross terms), dielectric, flexoelectric, carrier-transport, carrier-compressibility-and-coupling, and inertial contributions. The first, second, third, fifth and sixth lines of (39) are the discretization of (36); the third and fourth supply the piezoelectric cross terms that (37) needs; the seventh is (38), whose transport term enters with the opposite sign because each block is minus the residual of the weak equation and not the equation itself; the eighth is the compressibility and the added coupling, which appear in the carrier rows and, through $\widehat{\tau}$, in the mechanical rows as well. Equation (39) is the construction: each block is one quadrature of the listed integrands against the Legendre basis, so the entries can be regenerated from the stated integrands.

Each block takes one power of k_x from the test side and one from the trial side, but two of the nine vanish identically: $\mathsf{A}_{02}=\mathsf{A}_{20}=0$, because in each contracted pair $\eta_{ijk}\eta_{lmk}$ the components with $k=x$ carry at least one power of k_x and those with $k=y,z$ at most one, so that a factor with no power of k_x never meets one with two. The semi-discrete system is assembled as a polynomial of degree four in the propagation wavenumber k_x,

$$\Big[\sum_{r=0}^{2}\sum_{s=0}^{2}(-\mathrm{i}k_x)^r(\mathrm{i}k_x)^s\,\mathsf{A}_{rs}(\omega,k_y)\Big]\boldsymbol{q}=0, \tag{40}$$

where $\boldsymbol{q}$ collects all thickness degrees of freedom and the matrices A_{rs} contain the six blocks of (33); expanded power by power, (40) is (32). That degree is the order of the hyperelastic

operator and not of the interpolation: the three blocks that supply the cubic and quartic terms, A_{12}, A_{21} and A_{22}, are each proportional to ℓ^2 and vanish identically at $\ell = 0$, leaving the classical quadratic problem $\mathsf{A} = \mathsf{A}_{00} + k_x\mathsf{A}_{01} + k_x^2\mathsf{A}_{11}$; the two blocks A_{02} and A_{20} vanish for every ℓ.

The blocks also have the Hermitian structure one expects of an elastic problem with loss confined to the carriers: $\mathsf{A}_{12} = \mathsf{A}_{21}^{\mathsf{H}}$, $\mathsf{A}_{22} = \mathsf{A}_{22}^{\mathsf{H}}$ and $\mathsf{A}_{01} = \mathsf{A}_{10}^{\mathsf{H}}$ hold to 10^{-15}, so the cubic coefficient $\mathrm{i}(\mathsf{A}_{12} - \mathsf{A}_{21})$ is Hermitian, while A_{00} and A_{11} fail to be Hermitian by exactly the carrier-transport form of (39), which is the only non-Hermitian contribution in the construction.

The sizes and the scaling are recorded here, because they decide how the algebraic problem is solved. With n the polynomial degree, each field is expanded in $N = n + 1$ basis functions (50), so the three displacement components contribute $n_{\mathrm{u}} = 3N$, the potential $n_\varphi = N - 2$ on grounded faces (N on open-circuit faces) and the carrier $n_\mu = N$; the number of unknowns per thickness element is

$$n_{\mathrm{w}} = 3N + n_\varphi + n_\mu = 5n + 3 \text{ (grounded)}, \qquad 5n + 5 \text{ (open circuit)}, \tag{41}$$

so $n_{\mathrm{w}} = 113$ at the degree $n = 22$ used for the reported results, 93 at $n = 18$ and 168 at the reference degree $n = 33$ of fig. 9. The polynomial (40) is formally of degree $4n_{\mathrm{w}}$ in k_x, which would be 452 roots at $n_{\mathrm{w}} = 113$, but its leading coefficient A_{22} has rank n_{u} and not n_{w} (see (54)), so $2(n_{\mathrm{w}} - n_{\mathrm{u}}) = 2(n_\varphi + n_\mu)$ eigenvalues of the matrix polynomial are infinite and the finite spectrum contains

$$4n_{\mathrm{w}} - 2(n_{\mathrm{w}} - n_{\mathrm{u}}) = 2(n_{\mathrm{w}} + n_{\mathrm{u}}) \tag{42}$$

roots: 364 at $n = 22$, 300 at $n = 18$, 76 at degree four. That number is the state dimension, not twice it, and it is the reason the axial unknown u_3 enters the state (53) together with its derivative: the hyperelastic term is the only one containing k_x^2, and it is supported on the displacement slots alone. Half of the finite roots are outgoing and half incoming, but the split is not made by the sign of $\operatorname{Im} k_x$; the criterion is stated below (43). The Frobenius norms of the blocks at degree four and zero coupling are

$$\|\mathsf{A}_{00}\| = 1.89 \times 10^4, \quad \|\mathsf{A}_{01}\| = 5.67 \times 10^2, \quad \|\mathsf{A}_{11}\| = 2.35 \times 10^2,$$
$$\|\mathsf{A}_{12}\| = 4.04, \quad \|\mathsf{A}_{22}\| = 7.12 \times 10^{-1},$$

spanning four and a half decades. The spread is intrinsic to the problem: it is the $\lambda^2 c_{33}/c_{44} = 0.32$ of the gradient term against the $O(10^2)$ of the elastic term, multiplied by the $2j(j+1)(2j+1)$ growth of the stiffness matrix (51) with the mode number. Hence each block is equilibrated row by row before the eigenvalue problem is formed, and the conditioning quoted below is that of the equilibrated problem.

3.2 Modes, the free-edge system and the modal share

Two independent solution routes follow from (40).

The *temporal* route fixes a real k_x and solves the generalized eigenvalue problem in ω^2 obtained from (40) with the potential and the electrochemical potential eliminated. It produces the dispersion branches $\omega(k_x)$, the group velocity $v_g = c_s\,\partial\omega/\partial k_x$ by central differences along a branch-tracked sequence, and the temporal attenuation $\gamma = -2\operatorname{Im}\omega$. Branch tracking is done by a modal-assurance criterion on the mass-orthogonal displacements, so a curve can be followed through a crossing without jumping to a neighboring branch.

The *spatial* route fixes a real frequency and solves the quartic problem (40) directly for k_x. It is the route that is physically relevant at an edge, because reflection is a process at

fixed frequency, and it is the only route that gives the spatial attenuation $\alpha = 2|\operatorname{Im} k_x|$ used in figs. 1 and 6. The attenuation is taken from the complex root; it is not inferred from the imaginary part of $\sqrt{K^2 + k_y^2}$ at real K, which vanishes by construction.

The two routes are compared with each other in fig. 9(d) and with strong-form integrations of the reduced system. Both comparisons establish the consistency of the implementation, which is what they are used for; the independent checks on the physics are the exact limits and the finite-element plate.

At fixed ω the spatial problem returns $2(n_\mathrm{w} + n_\mathrm{u})$ finite eigenvalues $k_x^{(j)}$ and their states $\boldsymbol{y}^{(j)}$ (see (42)); the infinite ones are not states of the problem and hold no boundary data. Each is classified as outgoing or incoming by the sign of its decay and of its axial power,

$$\text{outgoing} \iff \operatorname{Im} k_x^{(j)} < -\tau \quad \text{or} \quad \left|\operatorname{Im} k_x^{(j)}\right| \le \tau \text{ and } \mathcal{P}_x^{(j)} < 0, \tag{43}$$

where $\tau = 2 \times 10^{-8} \max(1, |k_x|)$ and $\mathcal{P}_x$ is the thickness-integrated power flux (29). A purely real root is outgoing when it carries energy away from the edge, and an evanescent root is outgoing when it decays away from the edge. The number of outgoing roots must equal M, which is a structural property of the eigenvalue problem and not a tolerance. Varying τ from 10^{-9} to 10^{-6} times $\max(1, |k_x|)$ leaves the outgoing count (182 at degree 22), the identified backward root and R_B unchanged at both the operating point and the maximum.

The solution is written as the incident mode plus a combination of the outgoing modes,

$$\boldsymbol{y}(0) = \boldsymbol{y}_\mathrm{inc} + \sum_j a_j \, \boldsymbol{y}^{(j)}, \tag{44}$$

and the coefficients a_j follow from the homogeneous edge conditions (25) written as $\mathsf{B}\boldsymbol{a} = -\boldsymbol{b}_\mathrm{inc}$, with $\mathsf{B} \in \mathbb{C}^{m \times M}$. The system is square by construction, $m = M = n_\mathrm{w} + n_\mathrm{u}$ (see (56)), which at $n = 22$ is 182 rows against 182 columns; it is row- and column-scaled and solved through a singular-value decomposition, with the rank of B required to equal its column count at a cut-off of 10^{-12} of the largest singular value. A deficient system is reported as such and is not regularized. Because the system is square, a small residual is a statement about the conditioning of the edge operator and not about the physical admissibility of the edge conditions; the latter is argued from (25) independently. The condition number at the operating point is 1.4×10^2, and the scaled residual of (44) is below 10^{-12} for all rows used in this article.

The backward channel is the outgoing mode whose real wavenumber is closest to the target k_{Bx}, and its share of the incident power is

$$R_B = \frac{-\,P_B \, e^{2 \operatorname{Im} k_{Bx} L}}{P_\mathrm{inc} \, e^{2 \operatorname{Im} k_{ix} L}} = \frac{-\,P_B}{P_\mathrm{inc}} \, e^{2(\operatorname{Im} k_{Bx} - \operatorname{Im} k_{ix}) L}, \tag{45}$$

evaluated at the reference plane $x = -L$, $L = 10\,d = 2\,\mu\mathrm{m}$. The two exponentials are retained explicitly so that the ratio is measured at the same plane for both modes. Equation (45) defines a *modal share* of the outgoing power, one member of the partition of the incident power; the total propagating share, which is the quantity closest to a reflectance, is reported separately.

Evaluating (45) at the operating point takes four numbers from the solution and three steps, and the steps are the ones a reader can repeat by hand. The first is the power of the unit-amplitude backward mode at the edge, which the normalization makes independent of the degree:

$$\mathcal{P}_x = -0.0154248, \qquad |a_B| = 1.1184598, \qquad P_B^\mathrm{edge} = |a_B|^2 \mathcal{P}_x = -0.0192957,$$

the three numbers being read from the same $n = 22$ row. They are separately degree-dependent ($|\mathcal{P}_x|$ falls by 14 percent and $|a_B|^2$ rises by 16 percent between $n = 18$ and $n = 22$) and their product is not: 0.019296 at $n = 18$, 22, 26 and 30 alike, to six digits (see (58)). Quoting $\mathcal{P}_x$ or $|a_B|$ alone measures the normalization of the eigenvector, not the port. The second is the propagation to the reference plane, with the attenuation taken from the complex root,

$$\operatorname{Im} k_{Bx} = -5.2306 \times 10^{-6}, \quad \operatorname{Im} k_{ix} = +5.4027 \times 10^{-6} \text{ (per unit } d), \quad L = 10\, d,$$

$$P_B(-L) = -0.0192957\, e^{2(-5.2306\times 10^{-6})(10)} = -0.0192937,$$
$$P_{\text{inc}}(-L) = e^{2(5.4027\times 10^{-6})(10)} = 1.0001081,$$

where the incident mode is normalized to unit power at the edge. The third is the ratio,

$$R_B = \frac{-P_B(-L)}{P_{\text{inc}}(-L)} = \frac{0.0192937}{1.0001081} = 0.0192916,$$

that is 1.93 %. The two exponentials move the fourth digit in opposite directions: the backward mode decays and the incident mode grows as $x \to -L$, so the ratio changes by 2.1×10^{-4} relative to the unnormalized value $|a_B|^2|\mathcal{P}_x| = 0.0192957$, and it is that difference which (45) exists to account for. The same evaluation at the branch maximum gives 0.9429 (94.29 %), and the two values differ by the factor of forty-nine that section 4.3 attributes to the proximity of the branch to the coincidence condition.

A computed share is reported when it satisfies three conditions, none of which is a plotting threshold.

The backward mode is one of the outgoing modes, so its self-power cannot exceed the power available at the plane,

$$0 \le R_B \le 1 + \frac{|P_{\text{cross}}|}{P_{\text{inc}}}. \tag{46}$$

Across the reported solutions the interference term is between 10^{-6} and 10^{-2} of the incident power, so (46) is applied with a bound of 1.01. A row that violates it would assert that a single outgoing mode carries more power than the incident wave plus the interference, which no solution of the boundary-value problem can do; such a row indicates that the identified mode is not the continuation of the backward branch.

The reported quantity must be stable under refinement of the Galerkin degree. The peak study shows that both the operating-point value and the branch maximum change by less than 10^{-5} between degree 18 and degree 30.

R_B is a ratio of x fluxes. When the energy angle of a mode approaches $\pm 90^\circ$ its x flux vanishes and the ratio loses meaning even though the mode itself is perfectly well resolved. The flux fraction $|\cos\theta_E| = |P_x|/\sqrt{P_x^2 + P_y^2}$ is evaluated at each point, and the two values quoted in this article have $|\cos\theta_E| = 0.88$ and 0.97.

3.3 Validation

The implementation is checked in six independent ways, each of which compares a computed quantity with a closed-form or independently discretized one.

1. *Exact classical limit.* With $\ell = f = H = 0$ the plate supports exact shear-horizontal modes $\Omega^2 = (c_{66}/c_{44})K^2 + (j\pi)^2$ in scaled units; the spectral solution reproduces the first three of them to 10^{-9}.

2. *Exact strain-gradient limit.* With $\ell \neq 0$ the fundamental shear-horizontal branch has the closed form $\Omega^2 = (c_{66}/c_{44})K^2(1 + \lambda^2 K^2)$, reproduced to 10^{-10}.

3. *Strong-form propagator.* An eight-state classical sagittal system and a twelve-state gradient system are integrated directly as ODEs, with no spectral matrices. They reproduce the spectral frequencies to 10^{-10} for both the grounded and the open-circuit problem, and with nonzero f and H.

4. *Finite-element plate.* A C^1 finite-element plate formulation in the thickness direction (it keeps the modal x-dependence, so it tests the thickness discretization and not the two-dimensional corner; for finite-element analysis of second-order strain-gradient elasticity see [46]) converges monotonically to the spectral edge result at an observed order between 3.9 and 4.4, with an error below 3×10^{-5} on the finest mesh. This is the only check in the list that tests the edge problem; the dispersion checks do not.

5. *Two-dimensional finite-element solution.* The same variational problem is solved directly on the (x, z) cross-section, without modal x-dependence, at the operating point. The thickness basis is the spectral plate or the C^1 finite-element plate; in x a degree-4 B-spline basis is graded geometrically towards the edge, with a first cell of $0.02\,d$ in the reference case, and a mass-proportional absorbing layer of length $170\,d$ terminates the strip, because a conventional perfectly matched layer fails for backward waves. The incident mode enters as a scattered-field load, and the field on $-20d \leq x \leq -4d$ is fitted on the outgoing modes of section 3.2. The fitted backward share R_B agrees with the modal value 0.019292 to between -0.14 and $+0.42$ % over the variations in table 2. The test applies at the operating point only: at the fold the incident and backward modes coincide, so the modal fit that this comparison relies on is not defined there.

6. *Residuals of the strong form.* For each mode used, the momentum, Gauss, carrier-continuity, traction, double-traction and blocking residuals are below 10^{-8} in the normalized measure reported in fig. 9(c), and the scaled edge residual is below 10^{-12}.

7. *Natural boundary data: derived versus discrete.* The traction, double traction and corner line force of (26)–(27) are evaluated from the physical fields and compared, mode by mode, with the canonical flux rows that the solver actually imposes. The comparison is stated in (28); it closes to 1.4×10^{-13} for the incident guided mode at all degrees from 10 to 28 and for each quadrature rule tested, and it is the check that fixes the index placement and the sign of the corner term, so neither rests on the derivation alone. What the same comparison shows about the pointwise natural conditions (which a Galerkin statement imposes only weakly) is reported in section 5: the double traction is annihilated pointwise to 10^{-12}, whereas the traction leaves an interior residual of 9.4×10^{-4} of the incident scale at the corrected maximum (larger, 3.6×10^{-2}, at the operating point) that does not decrease further with degree, and the corner line force stands at 1.7×10^{-3} there and decays only algebraically. Those residuals are properties of the weak solution and not of the implementation, and they are reported with the other checks.

The reciprocity of the operator at $+k_y$ and $-k_y$ holds to 10^{-13}, which verifies the symmetry of the discretized bilinear form, including the added coupling. The seven comparisons above are the independent evidence for the implementation; the practical resolution at which they are carried out, the convergence of the reported quantities with degree and quadrature, and the conservation residual are stated in appendix B.

Table 2: Two-dimensional finite-element check at the operating point. Modal value $R_B = 0.019292$ (spectral plate; 0.019298 for the C^1 plate with 49 points). The reference case has strip length $200\,d$, absorbing layer $170\,d$ with strength 0.1, 500 uniform cells, first graded cell $0.02\,d$ and B-spline degree 4.

Variation of the reference case	R_B^{FE}	Difference (%)
Reference	0.019324	+0.17
Layer strength 0.05	0.019373	+0.42
800 uniform cells	0.019311	+0.10
B-spline degree 5	0.019265	−0.14
First cell $0.00125\,d$	0.019313	+0.11
C^1 plate (49 pts), first cell $0.00125\,d$	0.019320	+0.11

4 Numerical results

The operating point used throughout is $k_x d = 0.7362$, $k_y d = 0.1298$, $\omega = 7.6334\,c_s/d$, i.e. $f = 16.68\,\mathrm{GHz}$ for $d = 200\,\mathrm{nm}$, with the coupling set to zero unless a sweep says otherwise, and with grounded faces. The reference plane for each reported share is $L = 10\,d = 2\,\mu\mathrm{m}$ from the edge. Unless stated otherwise the Galerkin degree is $n = 22$.

4.1 Dispersion, wave speeds and the operating point

Figure 1 characterizes the guided branch that is later scattered at the edge. Panel (a) plots the frequency against $k_x d$ for the five coupling levels. The five curves superimpose to within the width of the line over the entire range $0.15 \le k_x d \le 1.05$, and the inset resolves their difference: the coupling lowers the frequency by at most 184 kHz, at $k_x d = 1.05$, where the frequency is 16.87 GHz, that is by 1.1×10^{-5} in relative terms; at the negative coupling the lowering remains below 92 kHz over the same range. This is the first indication, confirmed in section 4.6, that the added coupling is a perturbation of the dispersion and a modifier of the edge partition, and does not drive the branch.

Panel (b) shows three different velocities:

$$v_p = \frac{\omega}{|\boldsymbol{k}|} = \frac{\omega}{\sqrt{k_x^2 + k_y^2}}, \qquad \text{(phase speed, a magnitude)} \tag{47}$$

$$v_g = c_s \frac{\partial \omega}{\partial k_x}, \qquad \text{(group velocity, signed)}$$

$$v_E = \frac{\mathcal{P}_x}{E}, \qquad \text{(energy velocity, signed)}$$

with $\mathcal{P}_x$ from (29) and E the stored energy per unit length. The phase speed (47) is a positive magnitude and cannot change sign; it is plotted on the right-hand axis of panel (b) and it decreases monotonically from 105.8 km/s near the cutoff at $k_x d = 0.15$ to 20.0 km/s at $k_x d = 1.05$. The group and energy velocities are signed and are plotted on the left-hand axis: they run from −189 and −189 to +1136 and +1136 m/s, change sign just below $k_x d = 0.53$, and agree with each other to within 0.03 % at every sample; the group velocity is a symmetric difference of half-width 0.0025 in $k_x d$. The sign change of v_g and v_E marks the backward part of the branch. A backward branch exists whether or not an edge is present, and what the edge adds is the conversion into it: the branch provides a mode whose energy travels opposite to the phase, and that is the mode the edge subsequently selects.

Panels (c) and (d) report the linewidth and the attenuation. The linewidth ranges from 1.4 kHz to 0.27 MHz and grows monotonically with $k_x d$; it is smallest at $H = 0$ and rises with $|H|$, further for positive than for negative coupling. The attenuation $\alpha = 2|\operatorname{Im} k_x|$, obtained from the complex spatial root at real frequency and quoted in decibels of power as $4.343\,\alpha$, ranges from 0.11 dB/mm to 228 dB/mm and is positive everywhere on the branch. It is largest on the slow part of the branch, but the plotted grid of $\Delta k_x d = 0.03$ resolves that maximum only at its own sample, $k_x d = 0.54$: the fold has 43.5 dB/mm to 44.5 dB/mm at $H = 0$ (table 3), which a raster of that spacing cannot show.

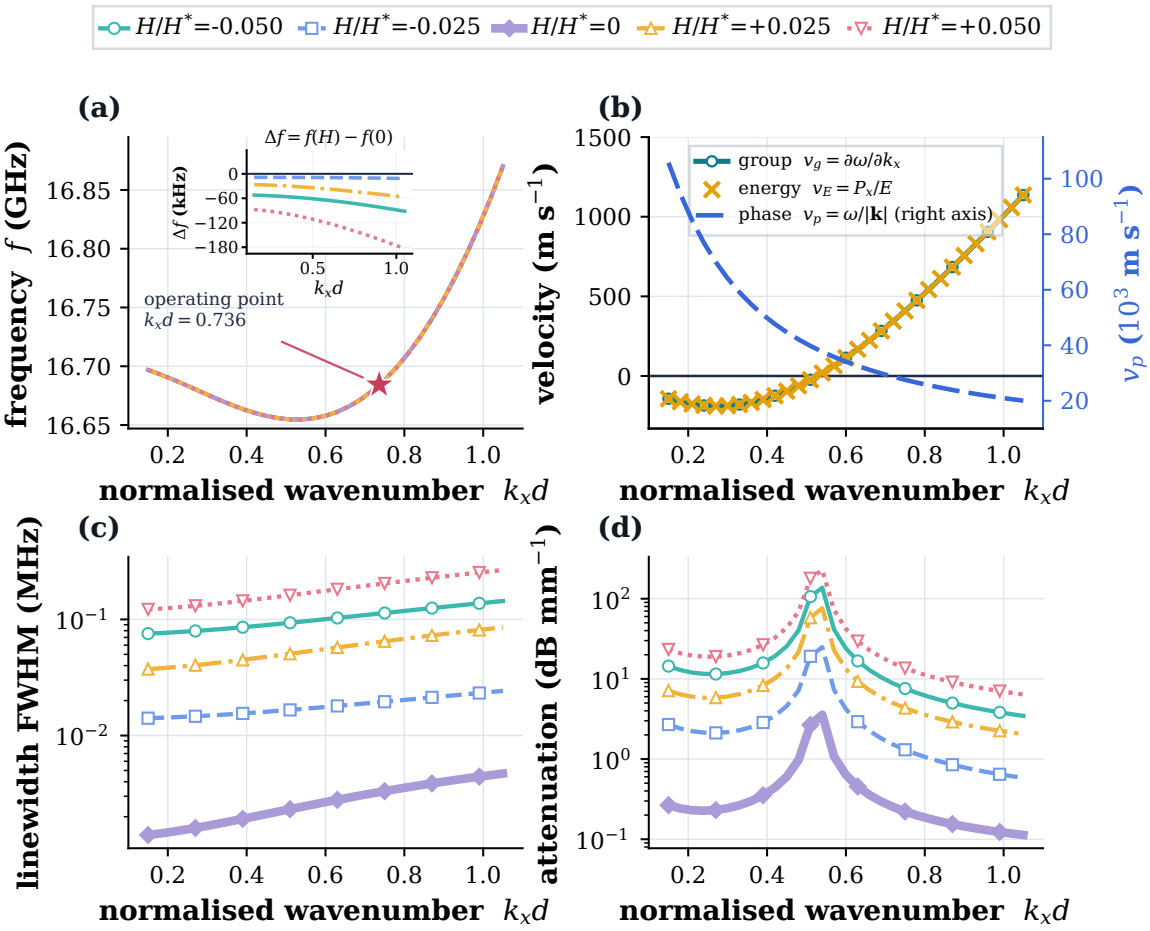


Figure 1: Guided branch at $k_y d = 0.1298$, $d = 200$ nm, $\ell = 50$ nm. (a) Frequency against $k_x d$, with the inset showing $\Delta f = f(H) - f(0)$; the operating point $k_x d = 0.736$ is marked. (b) Group, energy and phase speeds at $H = 0$. (c) Linewidth and (d) attenuation, $4.343 \times 2|\operatorname{Im} k_x|$ in dB mm^{-1}, for five coupling levels.

Before the numbers are read, the part of (6) exercised by the calculations should be stated.

Four sectors are active in each solve: the elastic, the strain-gradient, the piezoelectric and the carrier sector. The flexoelectric sector is active but subdominant. A single flexoelectric constant $f \equiv f_{3zz} = 1$ nC/m is retained. It lies at the low end of the range reported for ZnO [18, 44] and is an assumed value, taken independently of the crystal used for the elastic constants; it gives the dimensionless coupling $f = 2.56 \times 10^{-3}$ of table 1. The added carrier–strain-gradient coupling is swept over $|H| \le 2.0 \times 10^{-23}$ J m $= 0.022\, H_{\text{bound}}$, i.e. over the range allowed by (34).

The flexoelectric term was tested directly, by repeating the two solved configurations with f set to zero and to ten times the retained value. At the operating point the backward

channel is 0.0192916 with the retained f and 0.0192938 with $f = 0$, a change of $-0.012\,\%$; at $k_x d = 0.55$ it is 0.551910 and 0.551976, a change of $-0.012\,\%$. At ten times the retained value the changes are $-1.16\,\%$ and $-1.18\,\%$, so the dependence is smooth and quadratic in the coupling (a factor of 97 for a factor of ten), with no threshold.

Taken together, these numbers place the reported shares within one part in ten thousand of the shares of the *strain-gradient piezoelectric-semiconductor* specialization with the added carrier coupling. The flexoelectric term is present and contributes with the correct sign and magnitude, and it does not drive the effect. What the numbers support is a slow-mode edge effect of the strain-gradient plate, modulated by the carrier coupling, inside a flexoelectric-semiconductor formulation; they do not support the attribution of the negative reflection to the flexoelectric mechanism itself.

4.2 Dependence on carrier density, thickness and frequency

Figure 2 sweeps the two material families. Increasing the carrier density from 1 to $3 \times 10^{18}\,1/\mathrm{m}^3$ raises the linewidth (a) and the attenuation (b) by a factor of three each (the conduction loss is linear in the density over this range) and lowers the frequency only slightly. Panel (c) is the most informative: the backward channel at the reference plane is plotted against $k_x d$ for the five densities, and the five curves collapse onto one. The largest sampled value, $R_B \approx 0.552$ (0.5515 to 0.5523), is at $k_x d = 0.55$ for all densities; this raster of $\Delta k_x d = 0.05$ does not resolve the fold at $k_x d = 0.527$, whose maximum is given in section 4.3. The relative spread between the five curves is below $0.2\,\%$ over the full range. On this raster, and over this factor-of-three range, the carrier density thus sets the loss but not the partition of the scattered power (see section 4.8 for a wider range).

The mobility behaves in the same way (panel d): over the factor-of-three range 0.00875 to $0.02625\,\mathrm{m}^2/(\mathrm{V\,s})$ the linewidth changes by the same factor of three, while the backward share is unchanged to the same 0.2 percent. The two sweeps agree row by row because both quantities enter the conduction term through their product, while the density also enters the screening through $Q^2 = 5.6 \times 10^{-3}$, which is small here; the two are therefore equivalent at this Q and not interchangeable in general (see section 4.8 for larger densities). Both results are consistent with $Q = d/L_D = 0.075 < 1$: the plate is thinner than the Debye length, the carrier perturbation is not confined to a surface layer, and the density enters the problem mainly through the conduction loss.

Every row of this sweep satisfies the conservation bound (46); the curve in panel (c) extends to $k_x d = 0.9$, falling below 10^{-7} beyond $k_x d = 0.8$ and reaching between 3×10^{-12} and 1×10^{-11} at the last sample. This is the backward port ceasing to carry energy, not a numerical failure. Rows at that end have $|\cos\theta_E| \to 0$, and the flux fraction is evaluated at each point so that the loss of resolution is visible.

Figure 3 separates the two geometric families, which scale differently and must not be plotted on a common axis. Plate thickness is the dominant scale: across the family between 120 nm and 320 nm the branch lies between 10.5 GHz and 30.4 GHz, and the relative change is of order unity at all wavenumbers (panels a and c). The gradient length shifts the same branch by a smaller amount than the thickness (panels b and d): relative to $\ell = 50\,\mathrm{nm}$, a change to $\ell = 90\,\mathrm{nm}$ shifts the frequency by $2.0\,\%$ at $k_x d = 0.145$ and by $10.2\,\%$ at $k_x d = 1.00$, while $\ell = 35\,\mathrm{nm}$ shifts it by $-1.9\,\%$ and $-4.9\,\%$ at the same two points. The sign of the effect changes with ℓ across the reference value, so the panel is plotted as a relative change and not as an absolute frequency.

The distinction matters for the interpretation of the rest of the paper. A thickness change rescales the whole problem and is not a coupling effect; a gradient-length change alters the

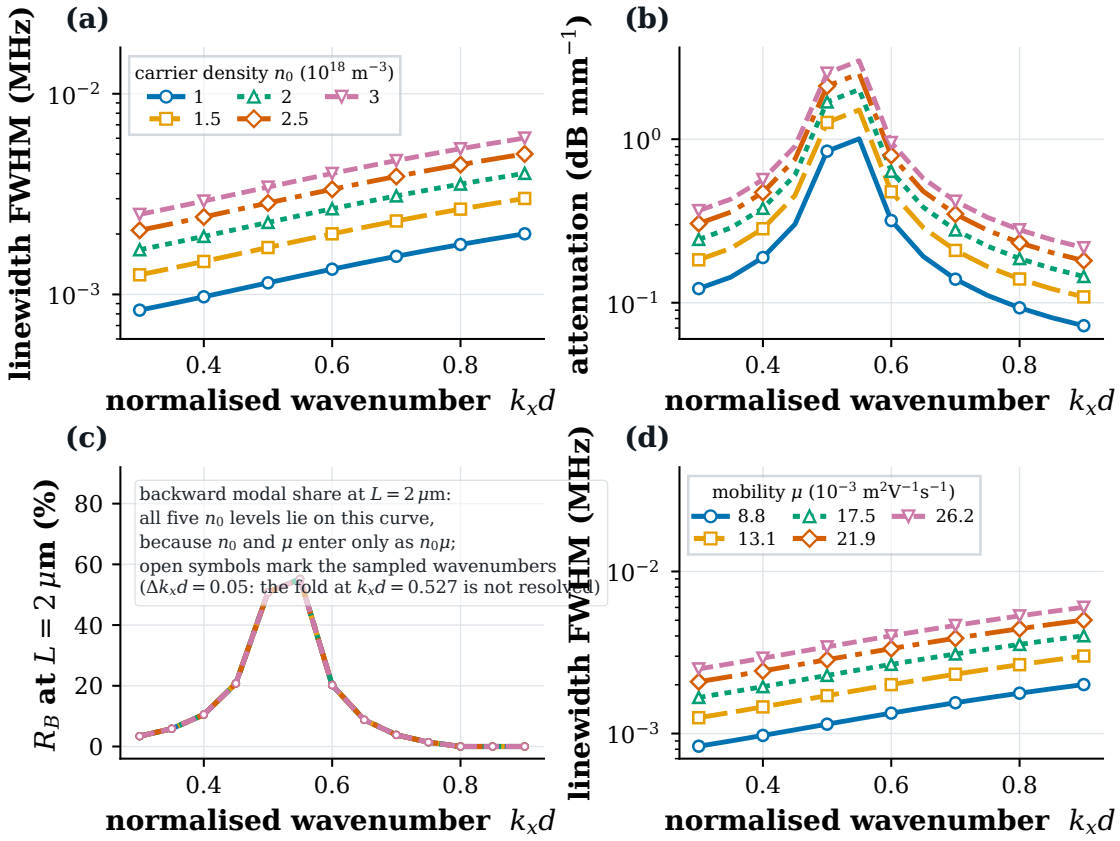


Figure 2: Carrier density and mobility at $H = 0$, $k_yd = 0.1298$, sampled every $\Delta k_xd = 0.05$. (a) Linewidth and (b) attenuation ($4.343 \times 2|\operatorname{Im} k_x|$) for five n_0; (c) backward share at $L = 2\,\mu$m; (d) linewidth for five mobilities.

relative weight of the hyperelastic and flexoelectric terms at fixed geometry and so is the parameter that belongs with H in a discussion of the added physics.

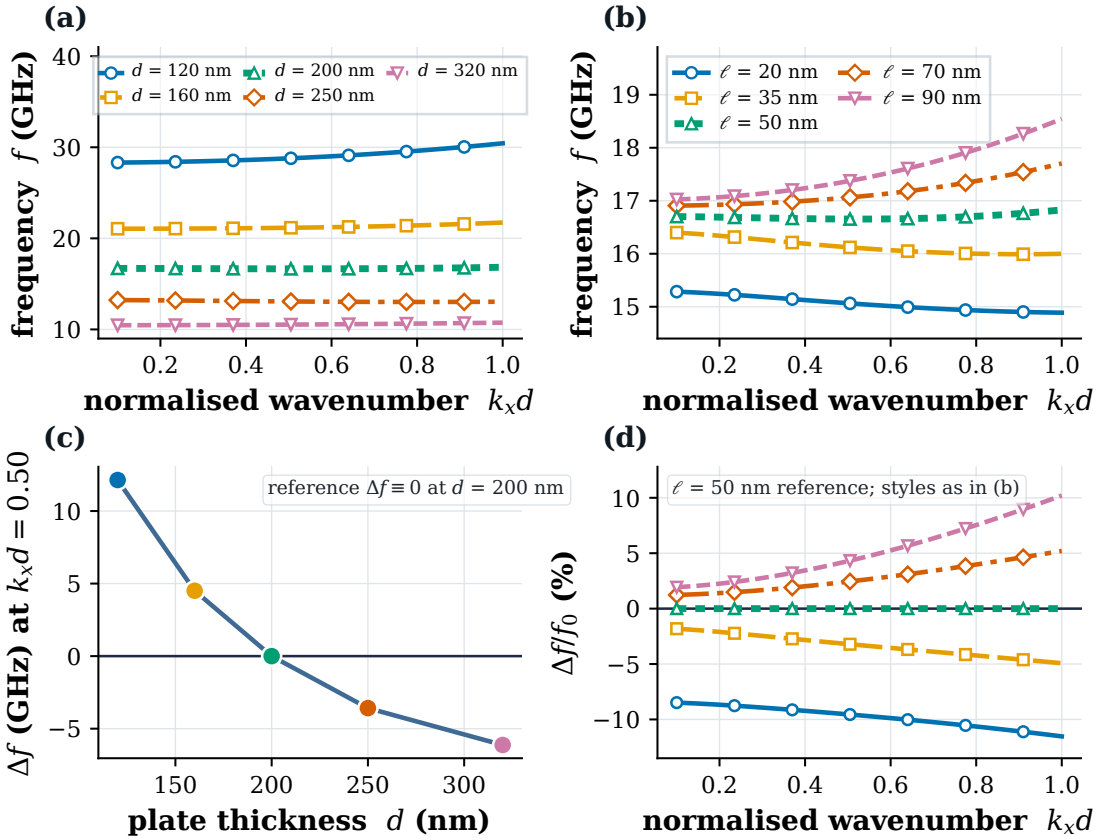


Figure 3: Thickness and gradient length at $H = 0$, $k_y d = 0.1298$. (a) Dispersion for five thicknesses ($\ell = 50\,\text{nm}$); (b) dispersion for five gradient lengths ($d = 200\,\text{nm}$), each the operating-point branch followed from $\ell = 50\,\text{nm}$; (c) Δf at $k_x d = 0.50$ against thickness; (d) relative frequency change for five gradient lengths.

Figure 4 sweeps the drive along the solved guided branch, not at a fixed frequency. This matters away from the operating point, where a fixed-frequency solve does not return the physical continuation of the branch and the energy angle collapses towards $-90°$; following the branch gives, for each row, a resolved triple of frequency, wavenumber and share. Two rasters are used: a coarse one over the full window and, at $H = 0$ only, a fine one of spacing $\Delta k_x d = 5 \times 10^{-4}$ across $0.520 \le k_x d \le 0.535$, because the top of the response is three orders of magnitude narrower than any raster that can be drawn on a single axis. The number of rows is a property of the grid.

The share along the branch is largest on its slow part. It is 0.0341 at $k_x d = 0.30$ ($f = 16.676\,\text{GHz}$), rises to 0.9429 at $k_x d = 0.52719$ ($f = 16.6547\,\text{GHz}$), and falls to 0.0142 at $k_x d = 0.75$ ($f = 16.688\,\text{GHz}$) (panel a). The response has a single peak over the entire window: the coarse raster finds no second local extremum, and the top of the peak is located by a refined search. Panel (b) shows the signed energy angle changing by 17° across the window and passing through its own extremum next to the maximum of R_B, so the two are not independent readings. Panel (c) plots the same data against the ratio k_{Bx}/k_{ix}: all five coupling levels collapse onto one master curve, and the conversion is largest where the backward wavenumber approaches the incident one, reaching $k_{Bx}/k_{ix} = 0.9960$ at the

maximum. The backward port is a propagating mode on the backward-slope part of the same branch, not an evanescent artifact, and the projection onto it becomes decisive where the two wavenumbers approach each other. Panel (d) shows why they do: the branch passes through a zero-group-velocity point at $k_x d \approx 0.527$, where $|v_g|$ falls below 1 m/s against -187 and $+409$ m/s at the two ends of the window, and the incident and backward wavenumbers close on one another there and open again beyond it.

In fig. 4 the five coupling levels separate only near the maximum (insets of panels a and c); the curves of panels (b) and (d) coincide on the scale of the full response. Panels (e) and (f) show that the energy angle changes by at most 7×10^{-2} degree, at the maximum, and the group velocity by less than 0.2 m/s. Every plotted row satisfies the conservation bound (46).

4.3 The branch maximum

The maximum of fig. 4 falls between two samples of the coarse raster, so it was solved again at six Galerkin degrees, each value being checked against the quadrature rule. At degree 30 the share is 0.019 292 at the operating point and 0.942 872 at the maximum. Between degrees 14 and 34 the relative change of the maximum stays below 10^{-5} and that of the operating point is 2.5×10^{-5}, the successive changes decreasing monotonically; with the degree fixed at 22 and the rule taken from 64 to 256 nodes the maximum changes by less than 10^{-8}. The edge system is well conditioned (condition number at most 1.9×10^{2}, scaled edge residual below 3×10^{-15}), and the values are resolved projections: the energy angles are $-13.9°$ and $-27.9°$, so the measured x flux is 97.1 % and 88.4 % of the total modal flux and neither value is an artifact of a near-grazing mode.

The quantity maximized is the one defined by (45): the modal self-power of the identified backward outgoing mode at the reference plane, divided by the incident power at the same plane. The other outgoing modes, the interference term and the dissipation are quoted separately and never folded into it. A coarse raster of $\Delta k_x d = 0.002$ over $0.465 \le k_x d \le 0.625$ (the whole window in which a backward port exists at this k_y) contains a single peak with its largest sample at $k_x d = 0.527$, and a plateau raster of 10^{-4} and a refined raster of 10^{-5} place the maximum at $k_x d = 0.52719$ with $R_B = 0.9428718$. The top is flat: every sampled wavenumber in $0.5267 \le k_x d \le 0.5276$ lies within 0.01 percentage point of the maximum, so the reported location is the center of that top, determined to within ± 0.0005. The incident wavenumber returned by the spatial problem is $k_{ix} d = 0.52821$, 0.19 % from the branch parameter, and no conclusion depends on the difference. The result is a *local maximum within the declared window*, not a global maximum over all k_y, thicknesses or material sets, none of which is searched here.

The maximum has a simple reading. At the fold the group velocity vanishes (+0.01 m/s against -187 and $+409$ m/s at the ends of the solved window) and the backward and incident wavenumbers differ by 0.4 %, so the two modes are nearly parallel and the edge conditions can be met with only a small admixture of the other outgoing modes; the projection onto the backward mode then takes almost all of the partition. Panel (c) of fig. 4 shows this: all coupling levels collapse onto one curve in k_{Bx}/k_{ix}, with the conversion maximal as the ratio approaches unity, and the conversion falls steeply on both flanks. A grid of $\Delta k_x d = 0.05$ would miss the maximum because the whole top of the response is 10^{-3} wide. The same fold makes the incident flux fall by a factor of 131 with respect to the operating point, so the partition is maximal where the transported power is minimal: that ninety-four percent of the energy is turned around would be true as a ratio and false as an energy budget, and only the ratio is claimed. The outgoing propagating modes carry all of the incident power to within

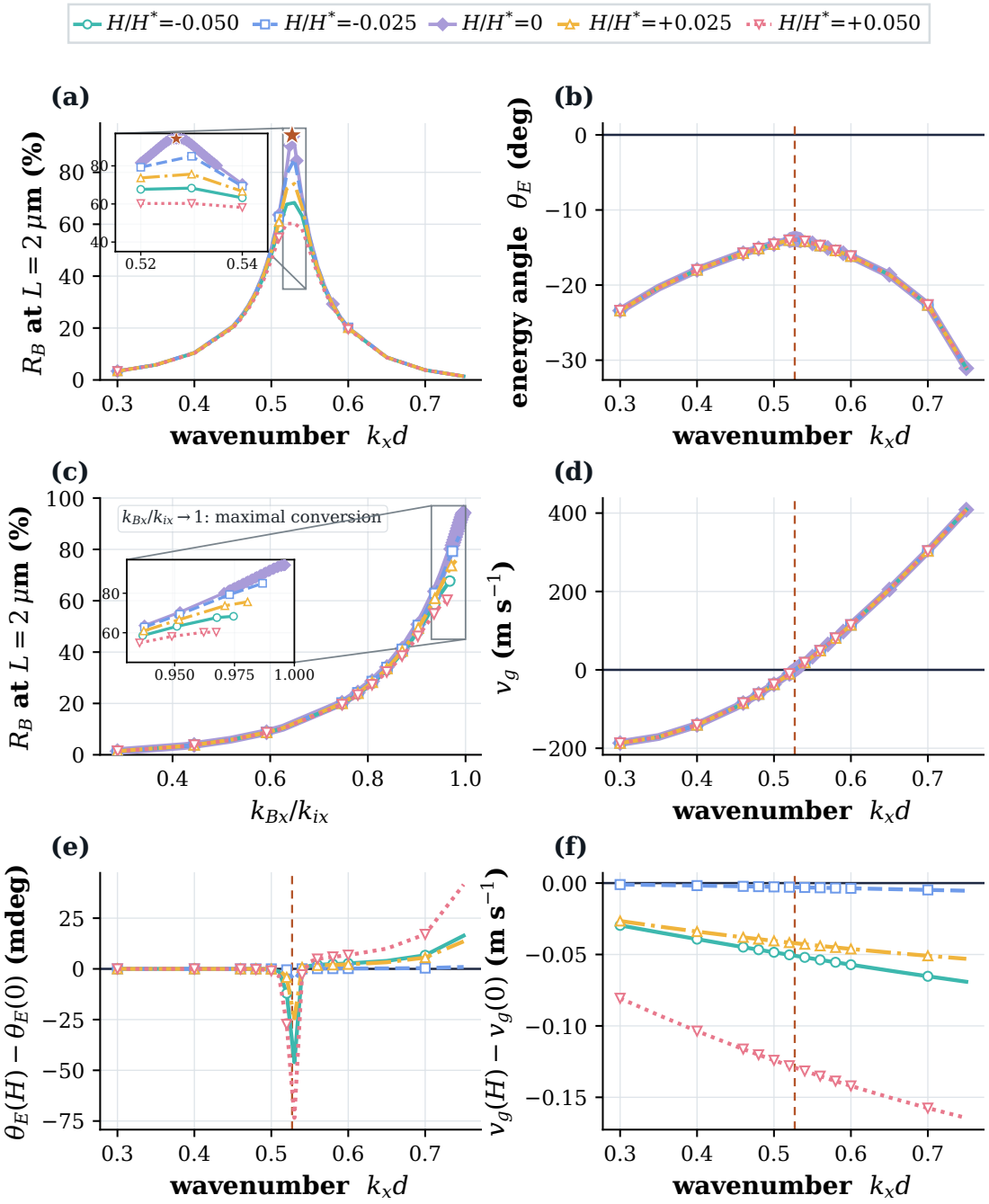


Figure 4: Backward conversion along the branch at $k_y d = 0.1298$, $L = 2\,\mu\text{m}$, with the frequency following the branch. (a) R_B against $k_x d$, maximum marked by a star. (b) Energy angle. (c) R_B against k_{Bx}/k_{ix}. (d) Group velocity. Insets magnify the region near the maximum; (e), (f) show the departure of (b) and (d) from $H = 0$.

0.1 % (section 4.4), so the plate is an almost perfect scatterer and the question is into which branch the power goes. A measurement of the total reflectance would see essentially unity everywhere on the branch; the negative reflection requires modal resolution, by a wavenumber filter or a time-of-flight separation of the branches.

4.4 Modal share and energy budget at the maximum

Three properties of the solution decide whether such a number can be read at face value.

The incident flux at the maximum is small, a factor 131 below its value at the operating point, so the fraction could in principle be a ratio of two nearly vanishing quantities. It is not: the value is reproduced to better than 10^{-5} across six Galerkin degrees, to better than 10^{-8} across four quadrature rules, and to 2.3×10^{-7} along three different tracking paths that reach the same wavenumber from below, from above and in a single step. A ratio of two independently vanishing numbers would scatter under all three.

The reference plane is moved from 0 to 20 thicknesses and the partition is rescaled by $\exp[2(|\operatorname{Im} k_{ix}| + |\operatorname{Im} k_{Bx}|)(L - L_{\mathrm{ref}})/d]$, which is the net attenuation of the two partner modes travelling in opposite directions. The rescaled values agree to 2×10^{-16}, so the number is a property of the edge and both attenuations are resolved and not assumed. The transform is applied identically to every result, and the edge-plane fractions of the search, of the matched controls and of the coupling study are reproduced by an independent re-evaluation to 10^{-10} relative or better; the plane at which a share is quoted is hence a reporting choice and not a second definition. The backward port is also unique: the second outgoing candidate lies $2.0|k_x|$ away, the condition number of the normalized edge matrix is 1.4×10^2, and the relative error bound of the projection is cond $\times\, \epsilon = 4 \times 10^{-14}$.

The incident and backward eigenvectors are nearly parallel, with overlap 0.999995, and their wavenumbers are separated by 1.45 times the modal decay rate: close, but resolved. The share also varies smoothly through the fold, taking the values 0.942 777, 0.942 869, 0.942 872, 0.942 866 and 0.942 818 at $k_x d = 0.5267$, 0.5270, 0.52719, 0.5274 and 0.5276 at degree 22, while the relative separation of the two wavenumbers stays within 0.2 percent of its value at the maximum. The partition rebuilt from the amplitudes and modal powers alone reproduces the computed value to machine precision, which is the direct statement that no eigenvector phase or scale enters the result. What the near-coincidence does change is interpretation, not accuracy: because the incident flux falls towards the fold, the fraction is anti-correlated with it across the window, so a maximal share goes together with a minimal transported power.

The maximum is a modal share and not a reflectance. At $k_x d = 0.52719$ the total outgoing propagating power is 1.00050 of the incident power and the interference correction at the reference plane is $+5.2 \times 10^{-4}$, so the backward mode carries 94.29 % of the incident power and 94.24 % of the outgoing propagating power. At this fold the interference term is sensitive at the level of 10^{-3} to the wavenumber at which the frequency is fixed (it is -8.0×10^{-4} at the neighboring sample $k_x d = 0.5275$, which has the same frequency to 10^{-8}), whereas the modal share changes by 2×10^{-5} relative. The modal share is 94.29 %; the energy statement is that the propagating outgoing modes carry the incident power to within 0.1 %. The two are different readings of the same solution.

The maximum must also be read with its own amplitude in mind. It lies on a zero-group-velocity fold, and a mode with vanishing group velocity carries vanishing flux: the incident modal power per unit normalization falls from 5.08×10^{-3} at the operating point to 3.87×10^{-5} at the maximum, a factor 131, while $|\operatorname{Im} k_{ix}| d$ rises to 1.0×10^{-3}. The fraction is the ratio of two small but resolved fluxes (well defined, unchanged by the incident normalization, and stable to 1×10^{-5} under degree and quadrature refinement), but it is not a statement that a

Table 3: The three reported configurations ($d = 200\,\mathrm{nm}$, $\ell = 50\,\mathrm{nm}$, $H = 0$, $k_y d = 0.1298$, $L = 10d$). R_B is a modal share, not a reflectance.

Quantity	Operating point	Maximum	$k_x d = 0.55$
frequency	16.684 GHz	16.6547 GHz	16.6550 GHz
longitudinal wavenumber $k_x d$	0.7362	0.52719	0.5500
incident wavenumber $\mathrm{Re}\, k_{ix} d$	0.7362	0.5282	0.5500
group velocity v_g	379 m/s	0.01 m/s	34 m/s
incident attenuation, $4.343 \times 2\lvert \mathrm{Im}\, k_{ix}\rvert$	0.235 dB/mm	43.5 dB/mm to 44.5 dB/mm	2.01 dB/mm
backward attenuation, $4.343 \times 2\lvert \mathrm{Im}\, k_{Bx}\rvert$	0.227 dB/mm	43.5 dB/mm to 44.5 dB/mm	1.96 dB/mm
backward share R_B	0.0192916	0.9428718	0.5519095
modal share at the edge plane	0.0193	0.9824	0.5529
energy angle θ_E	$-27.90°$	$-13.85°$	$-14.45°$
total outgoing propagating power	0.99980	1.00050	0.99878
interference $P_{\mathrm{cross}}/P_{\mathrm{inc}}$	-4.6×10^{-6}	$+5.2 \times 10^{-4}$	-1.03×10^{-3}
dissipation $P_{\mathrm{diss}}/P_{\mathrm{inc}}$	1.99×10^{-4}	2.78×10^{-5}	1.93×10^{-4}
edge-system condition number	$\leq 2 \times 10^{2}$	$\leq 2 \times 10^{2}$	$\leq 2 \times 10^{2}$
scaled edge residual	$\leq 3 \times 10^{-15}$	$\leq 3 \times 10^{-15}$	$\leq 3 \times 10^{-15}$

large amount of energy is turned around. Both readings are needed: the partition is maximal at the fold and the transported power is minimal there.

The three configurations are compared term by term in table 3. Every entry is evaluated at degree 30, except the attenuations, which are $2\lvert \mathrm{Im}\, k_x\rvert/d$ formed from the same spatial roots and quoted in decibels of power ($4.343\times$).

Figure 5 closes the energy identity (30) at the reference plane. At $H = 0$ the incident power normalized to unity is partitioned as follows: the identified backward channel carries 1.9292×10^{-2}, the remaining propagating outgoing modes carry 0.980505, dissipation removes 1.989×10^{-4}, and the interference term is -4.6×10^{-6}. The first two sum to 0.999796, the total outgoing propagating power; adding the dissipation leaves 5×10^{-6} of the incident power, which is the magnitude of the interference term to the six digits quoted. The interference term is defined as the remainder of the identity, so the sum closes by construction; the independent check is the balance of the flux at the reference plane against the edge flux and the conduction loss, which closes to better than 10^{-11} relative at each Galerkin degree from 14 to 34, five orders of magnitude below the 10^{-6} threshold plotted in panel (c). At the maximum the backward channel carries 0.942872 and the remaining propagating modes 5.76×10^{-2}, which together account for 1.00050 of the incident power; dissipation removes 2.78×10^{-5} and the interference term is $+5.2 \times 10^{-4}$. The same power balance closes there to 6×10^{-14}.

The evanescent spectrum does not belong to the transported budget. At the reference plane the entire evanescent content has decayed by a factor $\exp(-2|\mathrm{Im}\, k_x|L)$ with $|\mathrm{Im}\, k_x|L \gtrsim 10$, which leaves 4.8×10^{-28} of the incident power. Reactive accounting is meaningful only at the edge plane, where it is 4.4×10^{-6}, and even there it is not transported energy. For the same reason P_{cross} is an interference correction and not a channel, and the evanescent members of $\{P_j\}$ are reactive; the complete statement is the partition above, with the interference term quoted explicitly.

Panels (b) and (d) sweep the coupling at the *fixed* operating frequency $\omega = \omega_0$, so the incident wavenumber drifts with H along the branch instead of the frequency doing so. That is the response of the plate driven at one frequency; the branch-consistent comparison is the relevant one. At fixed frequency the backward channel changes from 1.9051×10^{-2} at $H/H^* = -0.05$ to 1.8876×10^{-2} at $H/H^* = +0.05$, a variation of 0.92 %; dissipation rises from 6.9×10^{-3} to 1.2×10^{-2}; the other propagating modes carry the balance. Panel (d) gives the same information as a relative change against the uncoupled baseline: the largest deviation is -2.15 % at $H/H^* = +0.05$. When instead k_x is held fixed and the frequency follows the branch (the branch-consistent comparison), the deviation at the same tabulated

extreme is $-1.21\,\%$: part of the fixed-frequency variation is the dispersion shift of H, which reaches 6.3×10^{-5} in ω.

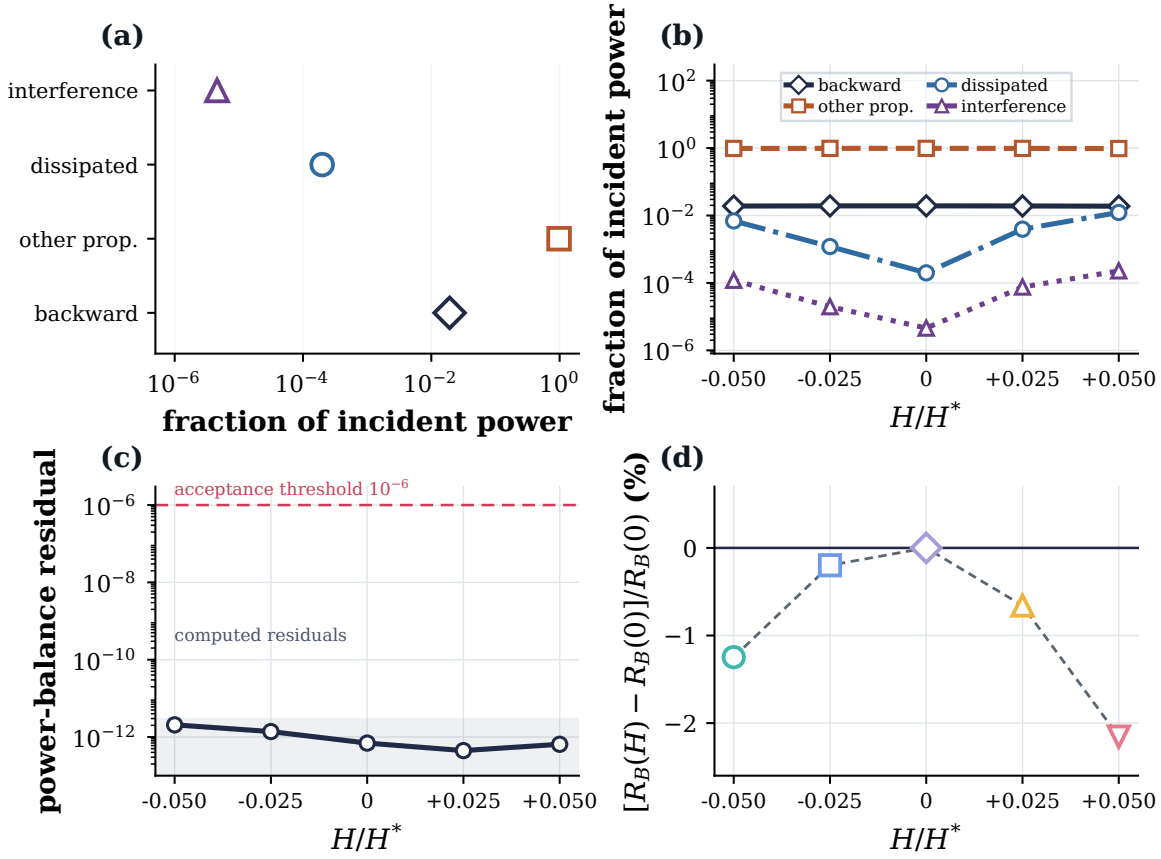


Figure 5: Energy budget at the operating point and the reference plane $x = -10\,d$; the interference term is shown as a magnitude. (a) Partition at $H = 0$. (b) The same fractions against H/H^* at fixed frequency. (c) Power-balance residual. (d) Relative change of the backward channel.

4.5 The backward channel

The partition quoted above was computed at $H = 0$, with the carrier–strain-gradient coupling switched off, and the contribution of that coupling is isolated by matched controls: each ingredient except the named one is held fixed (same k_x, same k_y, same n_0, same ℓ, same reference plane, same unit incident self-power), and the frequency follows the branch, because switching an ingredient off changes the dispersion and solutions taken at different frequencies of a different branch would not be comparable. Table 4 lists the outcome.

Three statements follow.

First, the effect is already present with $H = 0$: the same branch converts $1.93\,\%$ at the operating point and $94.29\,\%$ at the maximum. The strain-gradient piezoelectric-semiconductor model already contains the partition, and the coupling can only modify it.

Second, the size of the modification depends strongly on where the branch is sampled, and it is quantified. Turning H on at the tabulated extreme $|H| = 0.05H^*$ *reduces* the partition by $1.21\,\%$ at the operating point, by $10.17\,\%$ at $k_x d = 0.55$ and by $36.1\,\%$ at the maximum. The flexoelectric coefficient is neutral by comparison: at $f_0 = 1\,\mathrm{nC/m}$ it moves the partition by $-0.012\,\%$ at the operating point, by $-0.012\,\%$ at $k_x d = 0.55$ and by $0.0004\,\%$ at the

Table 4: Matched controls. The frequency follows the branch. Entries marked "sp" are specular self-powers, not backward conversion.

	Operating point ($k_x d = 0.736$)		Maximum ($k_x d = 0.52719$)		$k_x d = 0.550$	
Configuration	freq. (GHz)	R_B	freq. (GHz)	R_B	freq. (GHz)	R_B
full (f_0, $H/H^* = +0.05$)	16.6840	0.019 058	16.6546	0.602 292	16.6549	0.495 776
$f = f_0$, $H = 0$	16.6841	0.019 292	16.6547	0.942 872	16.6550	0.551 910
$f = 0$, $H/H^* = +0.05$	16.6840	0.019 060	16.6546	0.602 290	16.6549	0.495 822
$f = 0$, $H = 0$	16.6841	0.019 294	16.6547	0.942 878	16.6550	0.551 976
$\ell = 0$, $f = 0$, $H = 0$	16.2077	$0.075\,961^{\mathrm{sp}}$	15.9717	$0.133\,278^{\mathrm{sp}}$	15.9943	$0.124\,109^{\mathrm{sp}}$

maximum. The mechanism of the reduction is attenuation of both partner modes, whose decay rate rises from 1.0×10^{-3} to 8.5×10^{-3} per unit thickness at the maximum, so a larger part of the conversion is lost between the edge and the reference plane: referred to the edge plane, where both effects are evaluated at the same place, the reduction is 13.8 %, whereas it is 36.1 % at the reference plane, and the remaining 22.3 percentage points are propagation loss. At $k_x d = 0.55$ the same comparison gives 1.65 % at the edge plane, against −10.17 % at the reference plane: there the coupling slightly *raises* the edge-plane share, and the whole of the reference-plane reduction is propagation loss. The branch-consistent sweep is monotone in $|H|$: at the maximum $R_B = 0.9429$, 0.7564, 0.6023 for $H/H^* = 0$, +0.025, +0.05 and 0.8541, 0.6827 for −0.025, −0.05. When the extremum is re-searched afresh at each coupling level it moves to $k_x d = 0.5222$, 0.5292, 0.5272, 0.5279, 0.5240 for $H/H^* = +0.05$, +0.025, 0, −0.025, −0.05, and the largest attainable share falls to 0.6034 and 0.6832 at the two extremes, so the coupling lowers the upper bound as well as shifting the location, and the shift is not monotone in the sign of H.

Third, the gradient length fixes the frequency and the wavenumber of the fold; it does not create the backward interval. At the control frequencies of the last row of table 4, which lie on a different classical branch from the continuation followed below, the classical plate ($\ell = 0$) has no outgoing mode with positive axial wavenumber and negative axial power: the only negative-power outgoing mode is the specular reflection, whose wavenumber equals $-k_{x,\mathrm{inc}}$; those entries therefore carry the superscript "sp". Following the operating-point mode to $\ell = 0$ at fixed k_y, however, shows that the classical plate has a backward interval of its own. Along that path the frequency at $k_x = K_0$ falls from 16.684 GHz to 13.687 GHz, the fold moves from $k_x d = 0.527$ at $\lambda = 0.25$ to $k_x d = 0.824$ at $\lambda = 0$ (through 1.084 at $\lambda = 0.10$), and the group velocity is negative below the fold for every value of λ sampled.

The branch is tracked by modal overlap (at least 0.998 between neighboring steps), the group velocity is $\mathrm{d}\Omega/\mathrm{d}K$ by finite differences with $\Delta K = 0.01$, and the results agree to four digits between Galerkin degrees 18 and 24.

The edge problem was solved for the classical plate ($\lambda = 0$, no flexoelectricity, no coupling) at the fold of the tracked branch: at $k_x d = 0.824$ and 13.6855 GHz the backward mode carries 0.83620 of the incident power at the reference plane, and at $k_x d = 0.95$ (13.6891 GHz) it carries 0.95904. The spectral expansion at degrees 18 and 26 and the independent C^1 finite-element plate with 24 elements agree to 9×10^{-9} (first point) and 4×10^{-8} (second point), the power balance closes to 3×10^{-11} or better, and the scaled edge residual is below 3×10^{-15}. The large backward share is present in the classical plate before any strain-gradient length is introduced.

Taken together, the results read as follows. The negative reflection is a property of the

backward guided branch: on it the energy flux travels opposite to the phase, the free edge converts into it, and, at $\lambda = 0.25$, the conversion is largest where the branch group velocity vanishes; the gradient length determines where that fold lies. The carrier dynamics determine the attenuation of both branches and so the observed value at a finite reference plane. The added carrier–strain-gradient coupling acts through that attenuation: it leaves the partition almost untouched where the branch is fast, and removes a third of it at the slow point, because there the conversion is large and the added decay has the whole distance to the reference plane to act. It modifies the picture; it does not create it.

Figure 6 follows the two channels away from the edge. The backward coefficient decays with the observation distance L (panel a) as $\exp(-\alpha_B L)$ with $\alpha_B = 2|\operatorname{Im} k_{Bx}|$, and the incident power arriving at the edge obeys a survival factor normalized to its edge value, which cannot exceed unity (panel b); the minimum over the solved range, reached at $H/H^* = +0.05$, is 0.0356 at $L = 1000\,\mu\text{m}$, one millimeter from the edge.

Panels (c) and (d) give the attenuation length and the loss coefficient against the coupling. Both are obtained from the complex spatial roots, $\alpha = 2|\operatorname{Im} k_x|$, and neither is derived from a real square root. The attenuation length is largest at $H = 0$ ($\approx 18\,\text{mm}$) and falls by more than an order of magnitude at either extreme of the coupling range, to 0.54 mm at $H/H^* = -0.05$ and to 0.30 mm at $+0.05$, and the loss coefficient follows the same pattern with a minimum of 0.23 dB/mm at $H = 0$ and a maximum of 14.5 dB/mm at $H/H^* = +0.05$; the corresponding attenuation length falls from 18.5 mm to 0.30 mm, a factor of sixty-two. The incident and backward channels share the same attenuation only at $H = 0$ (0.235 and 0.227 dB/mm); away from it they separate strongly, reaching 8.08 and 11.5 dB/mm at $H/H^* = -0.05$. The coupling thus controls the loss of the plate much more strongly than it controls the partition of the scattered power, which is the opposite of what one might expect from the magnitude of the term (3) alone.

4.6 Coupling and the through-thickness flux

Figure 7 resolves the two-parameter dependence. Over the solved box $0.05 \le k_y d \le 0.20$ and $-0.05 \le H/H^* \le 0.05$, the backward channel ranges from 0.01546 to 0.02180. Panel (a) shows that the transverse wavenumber alone accounts for almost all of that range: the five coupling curves are displaced but their common slope is set by k_y. Panels (c) and (d) quantify the displacement: against H/H^* at fixed operating frequency the relative change is at most 2.48 %, and against k_y it is a slow drift that never exceeds the same bound. The ratio of the two sensitivities is the quantitative statement of this paper about the added coupling: k_y changes the backward channel by a factor of 1.41 across the box, while H changes it by at most 2.5 % at fixed frequency and by 1.21 % in the branch-consistent comparison. The coupling is a perturbation of the partition and not its organizing parameter. The comparison must be made at fixed k_x, because it is the branch and not the coupling that sets the partition; the two sensitivities do meet, however, at the branch maximum, where the branch-consistent coupling effect grows to 36.1 %. The k_y sweep of fig. 7 was solved at the operating point only.

Panel (b) draws the computed 5×19 grid cell by cell, with no interpolation beyond the computational box and no smoothing between the five coupling levels. Every row of the grid satisfies (46).

Figure 8 examines how the backward flux is distributed through the thickness. The plotted quantity is the x component of the backward Poynting flux, normalized by the incident flux, as a function of z/d; it is an evaluated field, not an interpolated surface.

The distribution is a signature of the mode and not of the edge. The x component changes sign inside the plate: it is positive near both free surfaces and negative in the core, reaching

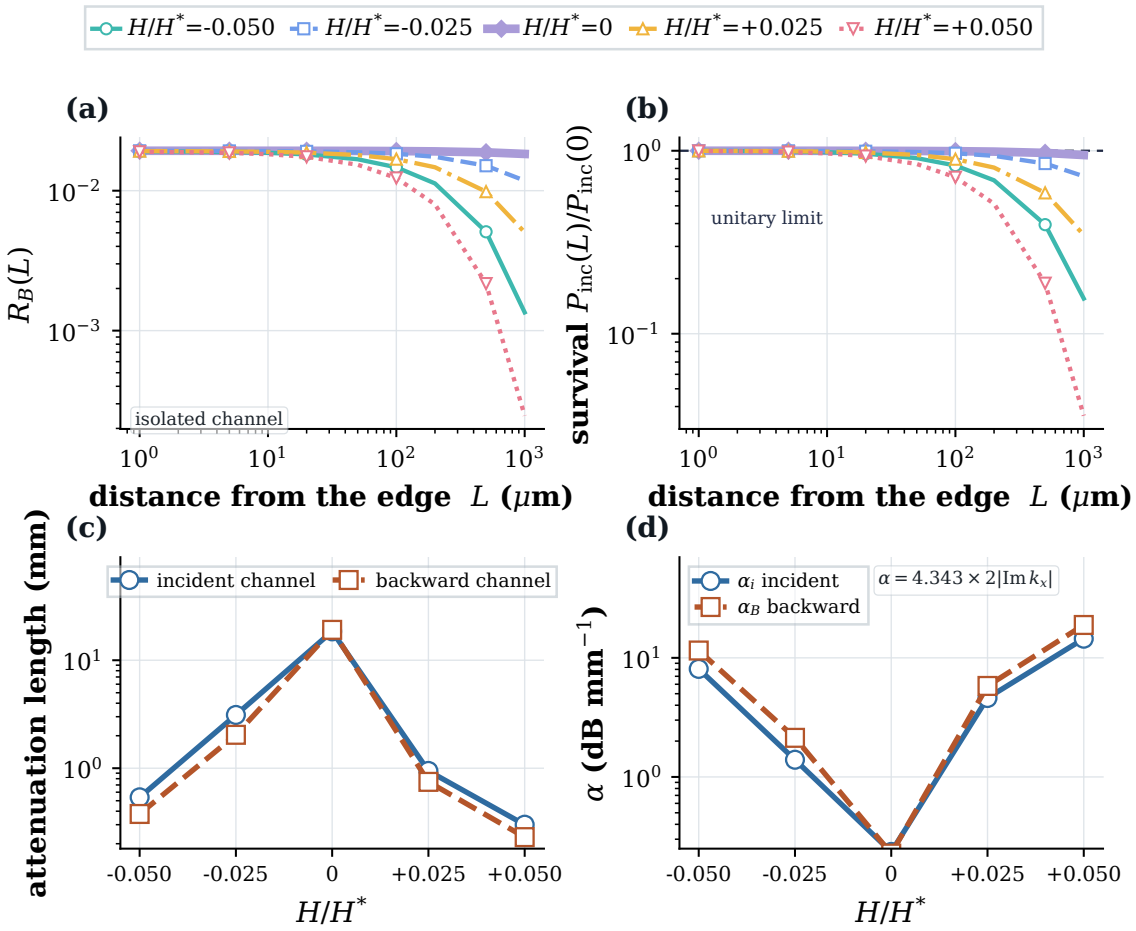


Figure 6: Distance, survival and attenuation at the operating point. (a) Backward channel against distance from the edge; (b) incident-power survival factor, normalized to its edge value; (c) attenuation length ($1/\alpha$ with $\alpha = 2|\operatorname{Im} k_x|$) and (d) loss coefficient in dB mm^{-1} ($4.343\,\alpha$) against H/H^*.

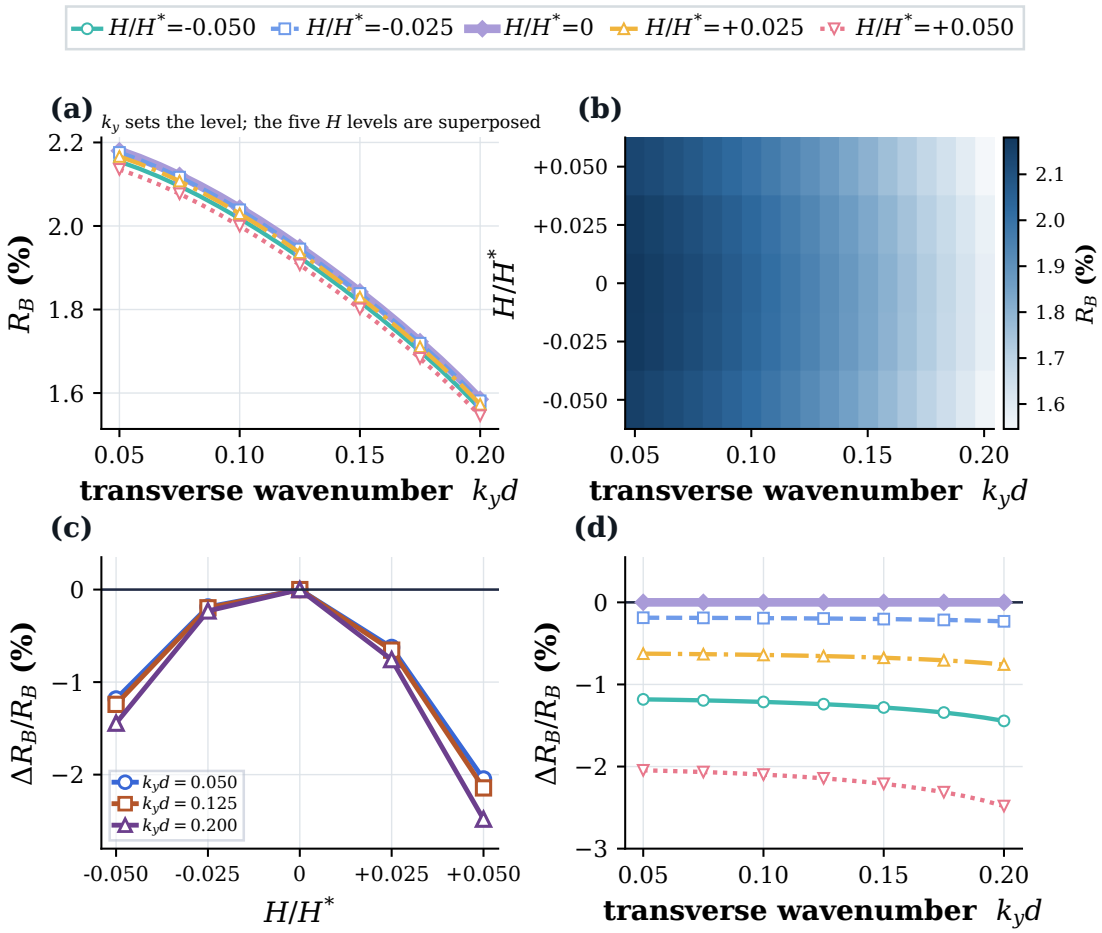


Figure 7: Coupling sensitivity at the operating frequency and wavenumber $k_x d = 0.7362$ (the operating point has $k_y d = 0.1298$). (a) R_B against $k_y d$ for five H/H^*; (b) measured $R_B(k_y d, H)$ grid; (c) relative change against H/H^* for three $k_y d$; (d) relative change against $k_y d$.

-0.0306 and $+0.0264$ in units of the thickness-integrated incident flux, while its thickness integral remains negative. The net backward power is carried by a mode whose local flux direction alternates across the thickness, which follows from the strain-gradient plate mode itself. The z component behaves differently: it is largest at the free surfaces and does not vanish there, so the surfaces carry a transverse flux even though both traction and double traction vanish on them. This is the configuration in which the reactive near field must be retained at the edge plane, and it explains why the energy budget separates the reactive content from the transported content instead of adding them.

Panel (c) gives the running thickness integral normalized to its own total, so the curve rises from zero to one across the plate for each coupling level and the five curves are indistinguishable. Panel (d) compares the shape of the backward and incident x fluxes at $H = 0$, both normalized by their own maxima; both profiles are symmetric about the mid-plane, the incident flux lies mostly near the two faces (55 % of $\int |P_x|\,\mathrm{d}z$ within $0.1\,d$ of them) and the backward flux does not (9 %).

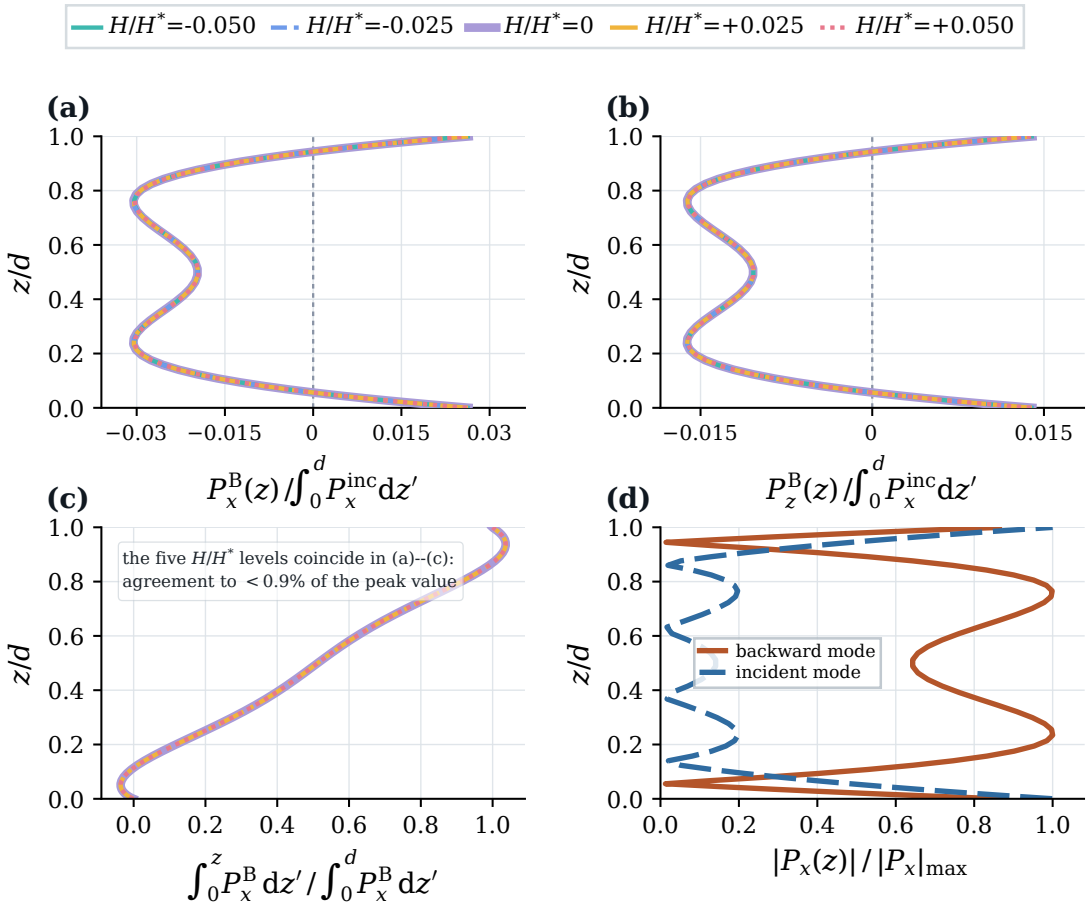


Figure 8: Through-thickness backward flux at the operating point and the reference plane, in units of the thickness-integrated incident flux $\int_0^d P_x^{\mathrm{inc}}\,\mathrm{d}z$. (a) $P_x^{\mathrm{B}}(z)$ and (b) $P_z^{\mathrm{B}}(z)$ for five H/H^*; (c) running integral of P_x^{B}; (d) backward and incident x fluxes at $H = 0$.

The coincidence of the five profiles in fig. 8(a)–(c) follows from the assembly and not from a plotting choice, and it can be quantified. The coupling enters through the two off-diagonal blocks that connect the out-of-plane displacement to the carrier, whose entries are

$$\left(\mathsf{A}_{u\mu}\right)_{ij} = h \int_{-1/2}^{1/2} b_i''(s)\, b_j(s)\,\mathrm{d}s, \tag{48}$$

because $\widetilde{\mu}$ contains $-h\,\partial_s^2 u_3$ and (10) is the relation that conveys the coupling; written in the coordinate $\zeta = 2s$ of the field equations the same term acquires a factor four, $4\partial_\zeta^2$, and the measure $\mathrm{d}s = \mathrm{d}\zeta/2$ is what keeps (48) free of it. Since b_i'' has degree $i-2$ and b_j has degree j, the matrix of (48) vanishes unless $j \le i-2$: it is lower triangular, not upper triangular, and its row norms 0, 0, 26.83, 91.65, 222.89, 450.87, 809.44 at degree six grow with the mode number as the second derivative does, the largest single entry being 580.5 in the last row. With $h = 1.06 \times 10^{-2}$ the largest entry of the assembled block is 6.13, which is 10.5% of the inertia scale $\omega^2 = 58.3$ that sets the through-thickness balance of the lowest rows, and four and a half orders of magnitude below the largest diagonal entry of the same rows, 2.16×10^5. That is a statement about the size of the coupling, not about a mode: the operator is row-equilibrated before the eigenvalue problem is formed, so no entry ratio is a frequency. The size of these entries sets the scale of the coupling in the assembled operator but is not used to derive the profile differences. Those are read from fig. 8(a)–(c): the flux profiles of the five levels differ by less than 0.9 % of the peak value, while the partition at the same coupling changes by 1.21 % at the operating point.

The same reading applies to the density sweep, where five carrier densities collapse onto one curve. The density enters the carrier sector only through the screening group $Q^2 = d^2 q^2 n_0/(\kappa_{33} k_B T)$, which multiplies the single potential block $\mathsf{A}_{\varphi\mu}$, and through the mobility in $\alpha_\mathrm{d} = \mu_e k_B T/(q\,d\,c_s)$; in the response the two appear as the conductivity $n_0\mu_e$, so at fixed $n_0\mu_e$ the partition is unchanged and the collapse reflects that product and not either parameter separately.

The ordering of these sensitivities is a property of the coupling and the operating point together: k_y changes the backward channel by a factor of 1.41 across the solved box while H moves it by 1.21 % at the operating point, but at the maximum the coupling moves the partition by 36.1 %, and the k_y sweep was not repeated there. Any statement about the added coupling must therefore be made at fixed k_y, and an experimental attempt to see it must control the transverse wavenumber much better than the doping or the mobility.

4.7 Validation and the near field

Figure 9 documents the numerical behavior of the edge solve. Panels (a) and (b) show the relative error of the backward channel and of the backward wavenumber against the degree-33 reference: the channel error falls from 2.6×10^{-5} at $n = 14$ to 1.6×10^{-6} at $n = 26$, while the wavenumber error remains at the 10^{-12} round-off floor over the full range, so the channel error is still decreasing at the highest degree shown. Panel (c) shows the scaled edge residual and the relative energy-balance error for each degree: the residual is 3×10^{-15}, six to seven orders below the 10^{-8} tolerance drawn in the panel, and the energy-balance error lies between 2×10^{-13} and 4×10^{-12}, one to three orders below the 10^{-10} line. Both are round-off in a relative measure, and the panel shows that neither drifts with the degree. Panel (d) compares the temporal and spatial routes for the same dispersion relation across the box; they agree to 2×10^{-5}.

The last panel is a check on the implementation, not a physical validation. The two routes discretize the same variational statement and differ only in which variable is used as the eigenvalue; agreement to 2×10^{-5} shows that the assembly, the boundary conditions and the elimination of the potential are implemented consistently.

The classical part of the edge solver was compared with the published free-edge calculation of Gérardin et al. [37]. The ZnO constants were replaced by those of their Duralumin plate ($\rho = 2790\,\mathrm{kg/m^3}$, $c_L = 6398\,\mathrm{m/s}$, $c_T = 3122\,\mathrm{m/s}$), with no piezoelectricity, strain gradient or carrier coupling, and the incident S_1 mode was taken at normal incidence ($k_y = 10^{-7}$

in the units of the solver). The backward branch appears at $f \times (2h) = 2.8727\,\mathrm{MHz\,mm}$, against 2.87 in [37]. The modal share $R_B = |r_{S_1|S_{2b}}|^2$ is 0.9988 at $2 \times 10^{-6}\,\mathrm{MHz\,mm}$ above that frequency, which is the perfect conversion of [37], and then falls to 0.797, 0.500 and 0.288 at 2.90, 3.00 and 3.10 MHz mm, that is $|r| = 0.893$, 0.707 and 0.537. Read from their Fig. 2(b) by eye, the corresponding values are about 0.9, 0.7 and 0.55, so the agreement is within the reading accuracy of the figure; no tabulated values are available for a closer comparison. Degrees 18 and 26 differ by less than 10^{-9} in R_B, the power balance closes to 4×10^{-11} and the edge residual is below 10^{-14}. This benchmark covers the classical elastic edge problem only; it does not test the piezoelectric, strain-gradient or carrier terms. The independent checks are the strong-form integrations and the C^1 finite-element plate, neither of which shares the spectral matrices with the production path.

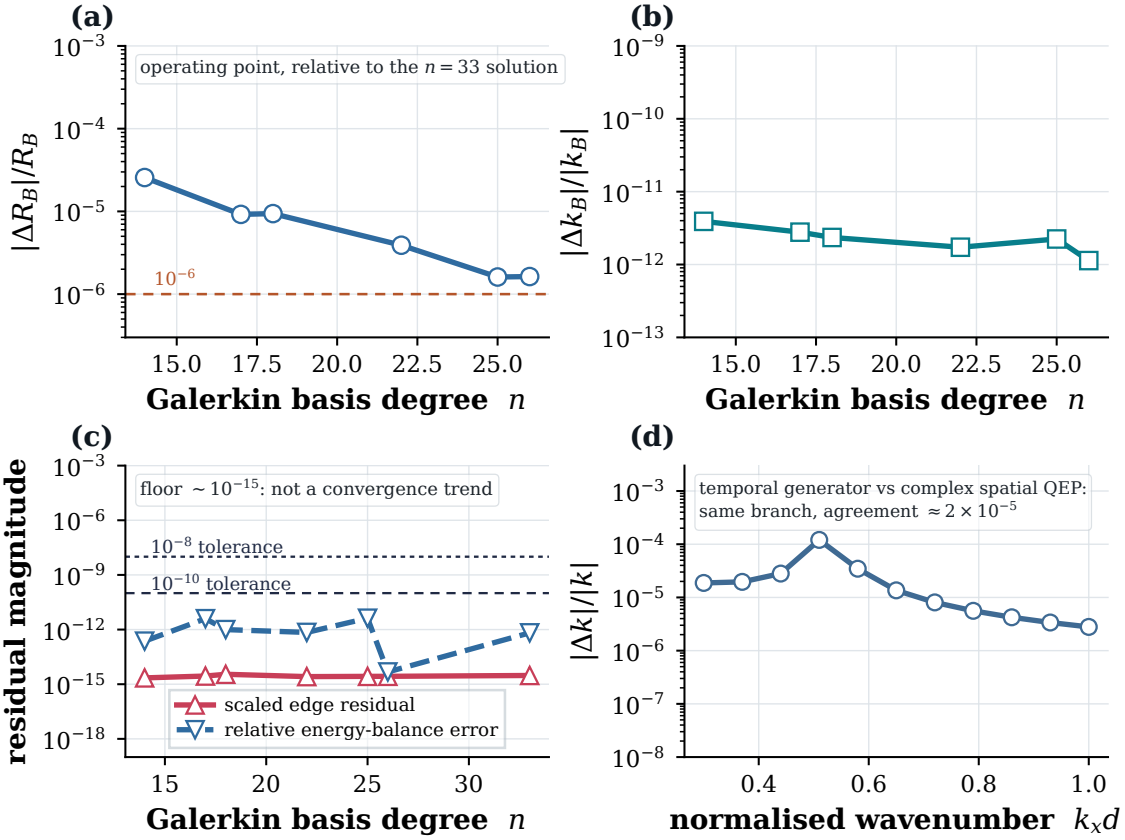


Figure 9: Convergence and consistency checks at the operating point. (a) Error of R_B and (b) of the backward wavenumber against the Galerkin degree, relative to the $n = 33$ solution; (c) edge residual and energy-balance error; (d) temporal-generator against spatial eigenvalue routes.

Figure 10 shows the solved field. Panels (a) and (b) show the incident and backward intensities on a shared logarithmic color scale over $x \in [-60, 1.5]\,\mu\mathrm{m}$ and $y \in [-45, 45]\,\mu\mathrm{m}$; the vertical line at $x = 0$ is the edge, the plate lies to its left, and the intensities are normalized to a common maximum so that the backward field is visibly weaker by one to three decades. The incident field approaches the edge at $+10.0°$ and the backward field leaves it at $-27.9°$, both measured from the thickness-integrated Poynting flux and from the x axis along the propagation direction (the $+x$ direction for the incident field, the $-x$ direction for the backward field), positive towards $+y$.

Panel (c) is a schematic, but its arrows are not drawn by hand: they are the solved

directions. The two wavenumbers point into the plate, $\boldsymbol{k}_i \propto (1, 0.176)$ and $\boldsymbol{k}_B \propto (1, 0.529)$, because k_y is positive and both axial wavenumbers are real and positive. The incident flux follows its wavenumber, $\boldsymbol{P}_i \propto (1, 0.176)$, whereas the backward flux is $\boldsymbol{P}_B \propto (-1, -0.529)$: the backward mode carries energy in $-x$ and in $-y$ simultaneously, so its energy transport is opposite in both components to its own phase direction, and the two flux arrows are the geometric content of the word "negative". Panel (d) shows the two components of the backward flux against $k_y d$; both are negative throughout the box, and both tend to a common small value as $k_y d \to 0.2$.

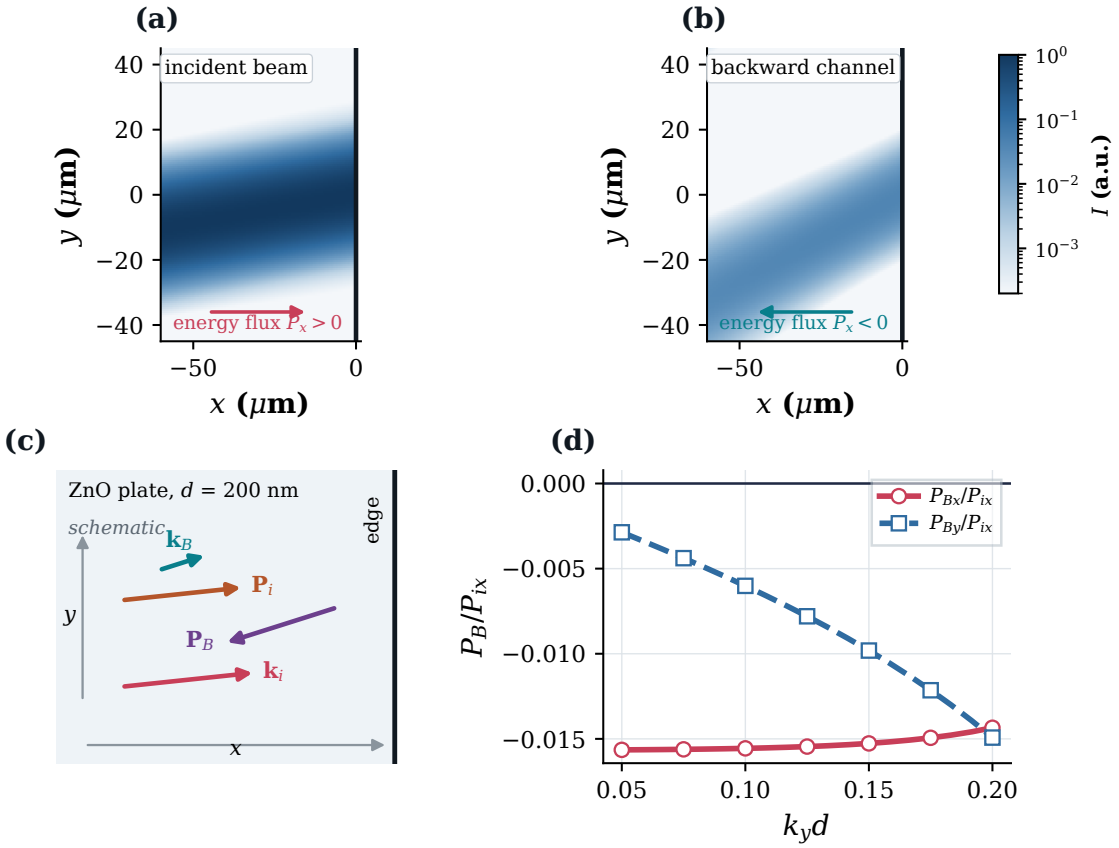


Figure 10: Field maps and flux directions at the operating point. (a) Incident and (b) backward intensity (arbitrary units, common logarithmic scale) for a Gaussian superposition in $k_y d$ of standard deviation 0.022 about 0.1298, edge at $x = 0$; (c) schematic of wavenumber and flux directions; (d) components of the backward flux against $k_y d$.

4.8 Carrier density and structural loss at the fold

The two sweeps above vary the carrier density over a factor of three and contain no loss other than carrier conduction and diffusion. Because the conversion is largest where the incident flux nearly vanishes, both limits were tested with the same branch-following definition of R_B ($L = 10\,d$, $k_y d = 0.1298$, $H = 0$).

The search for the fold was repeated at six densities from 2×10^{18} to $2 \times 10^{23}\,1/\mathrm{m}^3$, which takes $Q = d/L_D$ from 0.075 to 24. This range covers the densities of nominally undoped ZnO compiled in Table VII of [2], 8×10^{19}–$5 \times 10^{23}\,1/\mathrm{m}^3$, up to its upper end. The maximum falls from 0.943 through 0.828 to 0.545 over the first three densities (2×10^{18}, 2×10^{19},

$2 \times 10^{20}\,1/\mathrm{m}^3$), with the argmax inside the raster. From $2 \times 10^{21}\,1/\mathrm{m}^3$ upward the largest sampled value in a window of ± 0.03 about the fold lies on the edge of the window, so the entries (≥ 0.168, ≥ 0.043 and ≥ 0.174) are lower bounds and the maximum itself was not located. At the operating point, which was swept over a wider range than the fold search, the share stays between 0.0158 and 0.0410 over the whole range 2×10^{18} to $6 \times 10^{23}\,1/\mathrm{m}^3$, and its linewidth peaks near $2 \times 10^{22}\,1/\mathrm{m}^3$ at 8.7 MHz. The statement of section 4.2 that the density sets the loss but not the partition therefore holds for the range of fig. 2; at the fold the partition depends on the density through the loss.

For orientation, Zhang et al. [6] find for a simply supported plate under static load and in free vibration that the semiconductor coupling is active only within a window of initial electron concentration, which moves to larger concentrations as the plate becomes thinner; the concentration at which their damping is largest is $10^{18.6}$, $10^{19.6}$ and $10^{20.6}\,1/\mathrm{m}^3$ for $h = 30$, 3 and 0.3 µm. The present plate ($d = 200$ nm, $n_0 \leq 3 \times 10^{18}\,1/\mathrm{m}^3$ in the main sweeps) lies below these values, so the carrier results describe the weakly conducting side of that window. For shear-horizontal waves in ZnO plates, Li et al. [47] report that an initial density below $2 \times 10^{18}\,1/\mathrm{m}^3$ barely changes the cut-off frequencies and that above $10^{23}\,1/\mathrm{m}^3$ the plate responds as an elastic one, which is consistent with the density range scanned here. Dispersion and damping of Rayleigh-type waves in a piezo-thermo-electric semiconductor with fractional and nonlocal effects have also been computed [48].

The model contains no intrinsic elastic loss. To measure the effect of one, the elastic and gradient-elastic tensors were multiplied by $1 - \mathrm{i}\eta$ (time dependence $e^{-\mathrm{i}\omega t}$), a loss tangent η, i.e. a quality factor $1/\eta$, with the carrier terms unchanged. The form and the values of η are assumed; none is taken from a measurement. Viscoelastic loss of the Kelvin–Voigt type has been treated for nonlocal piezo-semiconductors elsewhere [49]. The frequency was scanned over 41 values within $\pm 3 \times 10^{-3}$ of the lossless fold frequency and the largest R_B is reported, so that each entry is a lower bound on the maximum over frequency. Refining the scan around its largest value in four successive steps, down to a window of 7.5×10^{-7} in relative frequency, changes the entries to 0.6003 and 0.1967, so the figures quoted below are converged in frequency. The scan shows that $\eta = 10^{-5}$ lowers the maximum from 0.943 to 0.600 and $\eta = 10^{-4}$ to 0.197, while $|\operatorname{Im} k_{ix}|d$ rises from 1.0×10^{-3} to 8.1×10^{-3} and 2.0×10^{-2}. This is the square-root response of the spatial roots to a small loss near a zero-group-velocity point. For $\eta \geq 10^{-3}$ the port selection finds no propagating incident mode in the scan window and the entries are not evaluated. The value 0.943 is therefore that of a plate whose only loss is carrier conduction, and it should be read as an upper bound for a real plate.

5 Conclusions

We have computed the scattering of a guided mode at the free edge of an n-type ZnO plate described by strain-gradient elasticity, piezoelectricity, flexoelectricity and linearized electron drift–diffusion, with one added carrier–strain-gradient coupling. The main findings are as follows.

1. At $k_x d = 0.736$ ($f = 16.68$ GHz) the backward branch carries 1.93 % of the incident power. Along the same branch the share rises to 94.29 % at $k_x d = 0.52719$ ($f = 16.6547$ GHz), where the group velocity vanishes. This is a local maximum along the examined branch, and it changes by less than 10^{-5} between Galerkin degrees 14 and 34.

2. These numbers are modal shares and not reflectances. At the maximum the incident

flux has fallen by a factor 131, so a maximal share occurs together with a minimal transported power.

3. The conversion belongs to the backward branch and not to the added coupling. The classical plate ($\ell = 0$) has its own backward interval, with its fold at $k_x d = 0.824$ ($f = 13.69\,\mathrm{GHz}$); for $\ell/d = 0.25$ the corresponding fold occurs at $k_x d = 0.527$ (16.65 GHz). The flexoelectric coefficient has only a minor influence on the partition in the range examined.

4. The carrier coupling acts through conduction loss. At $|H| = 0.05H^*$ it reduces the reference-plane share by 1.21 % at the operating point and by 36.1 % at the maximum, mostly through propagation loss between the edge and the reference plane.

5. The flux balance at the reference plane closes to better than 10^{-11} of the incident power.

6. At the fold the share is sensitive to loss: it falls from 0.943 to 0.545 when the carrier density rises from 2×10^{18} to $2 \times 10^{20}\,1/\mathrm{m}^3$, and to 0.600 and 0.197 for an added elastic loss tangent of 10^{-5} and 10^{-4}. The value 0.943 is therefore the lossless-elastic baseline of the model.

The model resolves, mode by mode, how a slow guided branch converts at a free edge when strain-gradient, piezoelectric, flexoelectric and carrier effects act together. It can serve as a basis for assessing edge-based wave conversion and loss in gigahertz piezoelectric-semiconductor plates, and its formulation extends to other piezoelectric semiconductors and edge configurations.

Data availability

All data generated or analysed in this study are included in this article; further numerical detail is available from the corresponding author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to check equations and numerical results, to verify bibliographic records and to edit the language. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

A Detailed mathematical derivations

This appendix contains the calculations that are used in the article but whose length would interrupt it: the derivatives of the enthalpy and the expanded field equations, the element matrices of the Legendre basis, the construction of the blocks (39) from the weak equations, the edge-matching system and the normalization of the modal power.

A.1 Derivatives of the enthalpy and the expanded field equations

Writing the enthalpy as the sum of its eight contributions, $w = \sum_\alpha w_\alpha$, the derivatives that give the constitutive relations are

$$\begin{aligned}
\frac{\partial w_{\mathrm{el}}}{\partial \varepsilon_{ij}} &= \tfrac{1}{2} c_{klij}\varepsilon_{kl} + \tfrac{1}{2} c_{ijkl}\varepsilon_{kl} = c_{ijkl}\varepsilon_{kl}, \\
\frac{\partial w_{\mathrm{grad}}}{\partial \eta_{ijk}} &= \tfrac{1}{2} A_{lmnijk}\eta_{lmn} + \tfrac{1}{2} A_{ijklmn}\eta_{lmn} = A_{ijklmn}\eta_{lmn}, \\
\frac{\partial w_{\mathrm{piezo}}}{\partial \varepsilon_{ij}} &= -e_{kij}E_k, \qquad \frac{\partial w_{\mathrm{flexo}}}{\partial \eta_{ijk}} = -f_{lijk}E_l, \\
\frac{\partial w_{\mathrm{diel}}}{\partial E_i} &= -\kappa_{ij}E_j, \qquad \frac{\partial w_{\mathrm{piezo}}}{\partial E_i} = -e_{ijk}\varepsilon_{jk}, \\
\frac{\partial w_{\mathrm{flexo}}}{\partial E_i} &= -f_{ijkl}\eta_{jkl}, \qquad \frac{\partial w_{\mathrm{carr}}}{\partial \Delta n} = a\,\Delta n, \\
\frac{\partial w_H}{\partial \Delta n} &= H_{jkl}\eta_{jkl}, \qquad \frac{\partial w_{\mathrm{es}}}{\partial \Delta n} = -q\varphi.
\end{aligned}$$

The symmetries $c_{ijkl} = c_{jikl} = c_{klij}$ and $A_{ijklmn} = A_{lmnijk}$ merge the two halves of each quadratic derivative; a tensor violating them would give a different stress from the same energy. Apart from the electrostatic term $-q\Delta n\varphi$, which is linear in Δn, w is homogeneous of degree two, and Euler's theorem gives

$$\varepsilon_{ij}\frac{\partial w}{\partial \varepsilon_{ij}} + \eta_{ijk}\frac{\partial w}{\partial \eta_{ijk}} + E_i\frac{\partial w}{\partial E_i} + \Delta n\frac{\partial w}{\partial \Delta n} = 2w + q\,\Delta n\,\varphi,$$

which gives the mode energy $w = \tfrac{1}{2}[\varepsilon : \sigma + \eta\dot{:}\tau - E\cdot D + \tilde{\mu}\,\Delta n] - \tfrac{1}{2}q\Delta n\varphi$ and is used in the energy budget.

The out-of-plane component of the momentum balance, in which all four sectors appear, is

$$\begin{aligned}
&\rho\omega^2\widehat{u}_3 + \partial_x\Big(2c_{44}\varepsilon_{xz} - e_{15}E_x\Big) + \partial_y\Big(2c_{44}\varepsilon_{yz} - e_{15}E_y\Big) \\
&\qquad + \partial_z\Big(c_{13}(\varepsilon_{xx}+\varepsilon_{yy}) + c_{33}\varepsilon_{zz} - e_{33}E_z\Big) - \partial_i\partial_k\tau_{3ik} = 0.
\end{aligned}$$

The flexoelectric and carrier couplings enter only through $\partial_z^2\tau_{3zz} \supset \partial_z^2(-fE_z + H\Delta n) = f\,\partial_z^3\varphi + H\,\partial_z^2\Delta n$, and the remaining terms of $\partial_i\partial_k\tau_{3ik}$ supply the fourth-order elastic piece in the combinations $\ell^2 c_{44}$ and $\ell^2 c_{33}$. Gauss's law (19) expands to

$$\begin{aligned}
&-\kappa_{11}\Big(\varphi_{,xx} + \varphi_{,yy}\Big) - \kappa_{33}\varphi_{,zz} + 2e_{15}\Big(\varepsilon_{xz,x} + \varepsilon_{yz,y}\Big) \\
&\qquad + e_{31}\Big(\varepsilon_{xx,z} + \varepsilon_{yy,z}\Big) + e_{33}\,\varepsilon_{zz,z} + f\,\eta_{zzz,z} = -q\,\Delta n.
\end{aligned}$$

A.2 Element matrices of the Legendre basis

Three coordinates appear in the formulation and they must be separated before any integral is written, because the orthonormality of the basis and the size of the element matrices both depend on which one is meant. The physical coordinate is $z \in [-d/2, d/2]$. The non-dimensional coordinate of the field equations is $\zeta = 2z/d \in [-1, 1]$, with $\mathrm{d}z = (d/2)\mathrm{d}\zeta$. The basis is generated on a third, the half-interval variable

$$s = \frac{z}{d} = \frac{\zeta}{2} \in \Big[-\tfrac{1}{2}, \tfrac{1}{2}\Big], \qquad \mathrm{d}s = \frac{\mathrm{d}z}{d} = \frac{\mathrm{d}\zeta}{2}, \tag{49}$$

and each element matrix in this appendix is an integral over s with the measure $\mathrm{d}s$. The basis is

$$b_j(s) = \sqrt{2j+1}\,P_j(2s), \qquad j = 0, \ldots, N-1, \tag{50}$$

with N the number of basis functions per field. Since $\int_{-1}^{1} P_i P_j \mathrm{d}\zeta = 2\delta_{ij}/(2j+1)$, the Gram matrix in the s measure is

$$\int_{-1/2}^{1/2} b_i b_j \,\mathrm{d}s = \frac{2j+1}{2} \int_{-1}^{1} P_i P_j \,\mathrm{d}\zeta = \delta_{ij},$$

the first factor being the Jacobian of $s = \zeta/2$ and the second the Legendre normalization. The same integral in the ζ measure is $\int_{-1}^{1} b_i b_j \mathrm{d}\zeta = 2\delta_{ij}$: the two differ by just the Jacobian, and the mass matrix that multiplies $\dot{\boldsymbol{q}}$ in the weak form is the identity only in the s measure. Hence all statements below are made with $\mathrm{d}s$, and the derivative in S_{ij} is $\mathrm{d}_s = \mathrm{d}/\mathrm{d}s$; in the ζ coordinate $\mathrm{d}_s = 2\mathrm{d}/\mathrm{d}\zeta$, which is the factor of two that appears in all strain-gradient entries.

The stiffness matrix is $S_{ij} = \int b_i' b_j' \mathrm{d}s$. For $i = j$ the chain rule gives $b_j'(s) = 2\sqrt{2j+1}\,P_j'(2s)$, so

$$S_{jj} = 4(2j+1) \int_{-1/2}^{1/2} \Big(P_j'(2s)\Big)^2 \mathrm{d}s = 2(2j+1)\, j(j+1), \tag{51}$$

where $\int_{-1}^{1} (P_j')^2 \mathrm{d}\zeta = j(j+1)$ is the classical companion of the Legendre equation and the factor 2 collects $4 \times \frac{1}{2}$. The weighted form of that identity is the one that follows directly from the equation, and it is derived here because this step is usually skipped. Multiplying $\Big[(1-\zeta^2)P_j'\Big]' = -j(j+1)P_j$ by P_k and integrating by parts, the boundary term carrying the factor $(1-\zeta^2)$ vanishes at both ends and

$$\int_{-1}^{1} (1-\zeta^2) P_j' P_k' \,\mathrm{d}\zeta = \frac{2j(j+1)}{2j+1}\,\delta_{jk},$$

which is the orthogonality of the derivatives under the weight $(1-\zeta^2)$; (51) is its unweighted companion and is used here only through the closed form it implies. The discrete matrix checks both, and so does an independent quadrature rule of a different order (appendix B): the assembled S_{jj} equals $2j(j+1)(2j+1)$ for every $j \le N-1$ at all degrees used in this article, to 2×10^{-11} or better in absolute value.

For $i \neq j$ the parity of $P_i' P_j'$ leaves $S_{ij} \neq 0$ only when i and j have the same parity, and

$$S_{ij} = 2\sqrt{(2i+1)(2j+1)} \int_{-1}^{1} P_i' P_j' \,\mathrm{d}\zeta,$$

which explains the structure of (52): the matrix is real and symmetric and it vanishes between basis functions of opposite parity, but the same-parity entries are not confined to a band. At $N = 6$ the entry $S_{15} = 22.978$ couples indices four apart, and at any N the row i has non-zero entries in each column $j \le i$ of the same parity, so the matrix is not tridiagonal and has no fixed bandwidth; what bounds it is the parity rule, not a band. At $N = 6$ the matrix is

$$\mathsf{S} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 12 & 0 & 18.330 & 0 & 22.978 \\ 0 & 0 & 60 & 0 & 80.498 & 0 \\ 0 & 18.330 & 0 & 168 & 0 & 210.60 \\ 0 & 0 & 80.498 & 0 & 360 & 0 \\ 0 & 22.978 & 0 & 210.60 & 0 & 660 \end{pmatrix}, \tag{52}$$

whose diagonal is (51) entry by entry. Equation (52) is the entire gradient-energy content of the discretization for a single field: it is real, symmetric and independent of the material, and each strain-gradient block of (39) is this matrix multiplied by an elastic modulus and by λ^2.

Two further element matrices are needed. The potential space of the grounded problem is spanned by

$$\psi_j(s) = b_j(s) - r_j\, b_{j+2}(s), \qquad r_j = \sqrt{\frac{2j+1}{2j+5}}, \qquad j = 0, \dots, N_\varphi - 1,$$

which vanishes at $s = \pm\frac{1}{2}$ for every j because $P_j(\pm 1) = (\pm 1)^j$ and r_j is exactly the ratio that makes the two terms cancel; $N_\varphi = N - 2$, since ψ_j needs b_{j+2} up to $j = N_\varphi - 1$, i.e. b_{N-1}. The two matrices that couple the charge are then distinguished by which index is the test index. Testing Gauss's law with ψ_i against the carrier basis b_j gives

$$\big(\mathsf{C}\big)_{ij} \equiv \int_{-1/2}^{1/2} \psi_i\, b_j \,\mathrm{d}s = \delta_{ij} - r_i\, \delta_{j,i+2}, \qquad i = 0, \dots, N_\varphi - 1, \quad j = 0, \dots, N - 1,$$

that is a unit matrix minus one band above the diagonal, with the coefficient r_i carrying the *row* index. The reciprocal matrix that appears in the carrier equation is its transpose,

$$\big(\mathsf{C}^{\mathsf{T}}\big)_{ij} \equiv \int_{-1/2}^{1/2} b_i\, \psi_j \,\mathrm{d}s = \delta_{ij} - r_j\, \delta_{i,j+2},$$

which has the band below the diagonal and the coefficient r_j on the column index. Both appear in (39): $-\mathsf{C}$ in the potential row and $-\mathsf{C}^{\mathsf{T}}$ in the carrier row, entering through the single term $-\int W_i^* W_j$ whose test and trial functions share the potentials. Writing r_j where r_i belongs, or transposing the two, gives a matrix that differs from the assembled one by 0.2981 in the largest entry at degree four, so the orientation is stated here and not left to the reader. Finally the dielectric matrix is $\kappa_{33}\int \psi_i'\psi_j' \mathrm{d}s + \kappa_{11} k_y^2 \int \psi_i \psi_j \mathrm{d}s$, with

$$\int_{-1/2}^{1/2} \psi_i'\psi_j' \,\mathrm{d}s = S_{ij} - r_j S_{i,j+2} - r_i S_{i+2,j} + r_i r_j S_{i+2,j+2},$$

in which the second and third terms are equal because S is symmetric, and the first term is not S_{ij} alone: the grounded basis removes a band of the stiffness exactly as it removes a band of the mass.

A.3 Construction of the blocks from the weak equations

Each of the nine blocks A_{rs} is the sum of the eight bilinear forms listed in (39). Written out with the operators of (33) and the integrals of the first subsection of this appendix, the block is

$$\begin{aligned}
\mathsf{A}_{rs} = {} & \underbrace{\big\langle \mathsf{B}_r, \mathsf{C}\mathsf{B}_s \big\rangle}_{\text{elastic}} + \underbrace{\lambda^2 \big\langle \partial_\zeta \mathsf{B}_r, \mathsf{C}\,\partial_\zeta \mathsf{B}_s \big\rangle}_{\text{gradient}} + \underbrace{\big\langle \mathsf{G}_r, \mathsf{e}\mathsf{B}_s \big\rangle + \big\langle \mathsf{B}_r, \mathsf{e}^{\mathsf{T}}\mathsf{G}_s \big\rangle}_{\text{piezoelectric}} \\
& - \underbrace{\big\langle \mathsf{G}_r, \boldsymbol{\kappa}\mathsf{G}_s \big\rangle}_{\text{dielectric}} + \underbrace{f\Big[\big\langle \mathsf{G}_r^{(2)}, \partial_\zeta \mathsf{B}_s^{(2)} \big\rangle + \big\langle \partial_\zeta \mathsf{B}_r^{(2)}, \mathsf{G}_s^{(2)} \big\rangle\Big]}_{\text{flexoelectric}} - \underbrace{\frac{\mathrm{i}\alpha_{\mathrm{d}}}{\omega} \big\langle \mathsf{M}\mathsf{G}_r, \mathsf{M}\mathsf{G}_s \big\rangle}_{\text{transport}} \\
& - \underbrace{\big\langle \mathsf{W}_r, \mathsf{W}_s \big\rangle}_{\text{carrier}} - \underbrace{\omega^2 \big\langle \mathsf{U}_r, \mathsf{U} \big\rangle \delta_{r0}\delta_{s0}}_{\text{inertia}},
\end{aligned} \tag{53}$$

with $\langle \boldsymbol{a}, \boldsymbol{b} \rangle = \int_{-1/2}^{1/2} \boldsymbol{a}^{\mathsf{H}} \boldsymbol{b}\, \mathrm{d}s$ in the measure of (49).

The powers of k_x are counted by the block indices (r, s) and by nothing else, and it is written out here because double counting is the usual error. The strain of a single mode is split once and for all as

$$\varepsilon_{ij} = \varepsilon_{ij}^{(0)} + \mathrm{i}k_x\, \varepsilon_{ij}^{(1)},$$

with both coefficient vectors independent of k_x:

component	$\varepsilon_{ij}^{(0)}$	$\varepsilon_{ij}^{(1)}$
ε_{xx}	0	b_j in the $\widehat{u}_1$ slot
ε_{yy}	$\mathrm{i}k_y b_j$ in the $\widehat{u}_2$ slot	0
ε_{zz}	b'_j in the $\widehat{u}_3$ slot	0
ε_{yz}	b'_j in $\widehat{u}_2$, $\mathrm{i}k_y b_j$ in $\widehat{u}_3$	0
ε_{xz}	b'_j in the $\widehat{u}_1$ slot	b_j in the $\widehat{u}_3$ slot
ε_{xy}	$\mathrm{i}k_y b_j$ in the $\widehat{u}_1$ slot	b_j in the $\widehat{u}_2$ slot

no component contains more than one power of $\mathrm{i}k_x$, so the split terminates at first order. The strain gradient is split the same way, with ∂_x acting on the test factor and on the trial factor separately, and the block is defined as the coefficient of $(-\mathrm{i}k_x)^r(\mathrm{i}k_x)^s$ in (53):

$$\mathsf{A}_{rs} = \Big\langle \varepsilon^{(r)}, \mathsf{C}\varepsilon^{(s)} \Big\rangle + \lambda^2 \Big\langle \partial_z \varepsilon^{(r)}, \mathsf{C}\partial_z \varepsilon^{(s)} \Big\rangle + \cdots ,$$

the remaining terms of (53) being split in the same way. Two consequences follow in this language. First, the leading coefficient is the gradient energy of the axial part of the strain alone,

$$\mathsf{A}_{22} = \lambda^2 \Big\langle \varepsilon^{(1)}, \mathsf{C}\varepsilon^{(1)} \Big\rangle \neq 0, \tag{54}$$

which is where the quartic term and the four wave families come from; its rank is n_{u} and not n_{w}, because $\varepsilon^{(1)}$ is supported on the displacement slots only. Secondly, no strain component contains two powers of $\mathrm{i}k_x$, which is why $\mathsf{A}_{02} = \mathsf{A}_{20} = 0$ and the polynomial is a quartic in the (r, s) bookkeeping of (40), not a sextic: the k_x^4 term needs one factor from each side of the gradient energy and cannot be produced by a single factor carrying two.

The same bookkeeping fixes where the flexoelectric constant can appear. The only non-zero component of f_{ijkl} used in this article is f_{3333}, which pairs the field E_z with the gradient component $\eta_{zzz} = \partial_z \varepsilon_{zz}$; neither contains a power of k_x, since $E_z = -\partial_z \varphi$ and ε_{zz} is axial, so f enters A_{00} and no other block. That is a statement about the operator, and it explains why a non-zero f can change the elastic-like term and the double traction without touching the quartic coefficient A_{22}: the two constants sit in different blocks, and neither of them is a k_x power.

A.4 The edge-matching system

At the edge $x = 0$ the conditions (25) are imposed weakly, one row block per condition and each on the test functions of the field it multiplies. The normal there is the x direction, not the z direction: the generalized traction $\bar{\sigma}_{1j}$ on the $n_{\mathrm{u}} = 3N$ displacement test functions, the double traction $\tau_{1j1} \equiv \tau_{jxx}$ on the same n_{u} test functions, the normal electric displacement D_x on the n_φ potential test functions and the normal carrier flux J_x on the n_μ carrier test functions. The face quantities $\bar{\sigma}_{iz}$, τ_{izz} and D_z of (22)–(24) belong to $z = \pm d/2$; they are the natural conditions of the expansion in z and are not the edge conditions. The number of rows is thus

$$m = 2n_{\mathrm{u}} + n_\varphi + n_\mu,$$

with the double-traction block present whenever the hyperstress is non-zero. At the edge only the axial strain gradient remains:

$$\tau_{ijx} = \ell^2 c_{ijkl}\eta_{klx}. \tag{55}$$

The two gradient couplings of (14) act on the normal-derivative component: f pairs E_z with η_{zzz} and H pairs Δn with the same component, so neither appears in (55). The block exists if and only if $\ell \neq 0$, independently of f and of H: it is present in cases A, B and D below, and it is empty in the classical limit $\ell = f = H = 0$, where $\tau_{ijx} \equiv 0$ and the double-traction statement degenerates to $0 = 0$ and does not give a boundary condition. The case $\ell = 0$ with $f \neq 0$ retains a hyperstress but has no axial trace to pair it with, which is why it is not a case of this model. Between those cases the edge problem changes order, and with it the state:

case	parameters	τ_{ijx}	N_s	m	double traction
A	$\ell \neq 0$, $f \neq 0$, $H \neq 0$	$\ell^2 c_{ijkl}\eta_{klx}$	$2n_\mathrm{w} + 2n_\mathrm{u}$	$2n_\mathrm{u} + n_\varphi + n_\mu$	required
B	$f = 0$, $\ell \neq 0$	$\ell^2 c_{ijkl}\eta_{klx}$	$2n_\mathrm{w} + 2n_\mathrm{u}$	$2n_\mathrm{u} + n_\varphi + n_\mu$	required
C	$\ell = 0$, $f \neq 0$	0 (no axial gradient)	$2n_\mathrm{w}$	$n_\mathrm{u} + n_\varphi + n_\mu$	*not available*
D	$H = 0$, $\ell \neq 0$	$\ell^2 c_{ijkl}\eta_{klx}$	$2n_\mathrm{w} + 2n_\mathrm{u}$	$2n_\mathrm{u} + n_\varphi + n_\mu$	required
E	$\ell = f = H = 0$	0	$2n_\mathrm{w}$	$n_\mathrm{u} + n_\varphi + n_\mu$	absent (identically zero)

Case C is not a case of the present model: the hyperstress would be non-zero while the second-order state of case E, which has no $\partial_x u$ trace to pair the double traction with, is the only one available at $\ell = 0$. Thus the formulation admits $\ell = 0$ only in the classical limit $f = H = 0$, and states that restriction explicitly; it does not assume that $\ell = 0$ implies a second-order model. No number in this article depends on case C: every reported run has $\ell \neq 0$.

With the edge ansatz (44) and the modal state $\boldsymbol{y}^{(j)} \in \mathbb{C}^{N_s}$ (the n_w displacement and potential coefficients, the n_u axial traces, and the n_w and n_u conjugate rows), the system is

$$\boldsymbol{y}(0) = \boldsymbol{y}_\mathrm{inc} + \sum_{j=1}^{M} a_j\, \boldsymbol{y}^{(j)}, \qquad \mathsf{B}\boldsymbol{a} = -\boldsymbol{b}_\mathrm{inc},$$

where the boundary operator acts on a *state* and the boundary matrix on the modal index,

$$\mathsf{R} = \begin{pmatrix} \mathsf{R}_\Sigma \\ \mathsf{R}_\tau \\ \mathsf{R}_D \\ \mathsf{R}_J \end{pmatrix} \in \mathbb{C}^{m\times N_s}, \qquad \mathsf{B} = \left[\mathsf{R}\boldsymbol{y}^{(1)} \;\cdots\; \mathsf{R}\boldsymbol{y}^{(M)}\right] \in \mathbb{C}^{m\times M}, \qquad \boldsymbol{b}_\mathrm{inc} = \mathsf{R}\,\boldsymbol{y}_\mathrm{inc} \in \mathbb{C}^{m},$$

so the three objects have three different second dimensions and only the last two agree. They do agree, identically, not by accident: the number of decaying states is half the state dimension, $M = N_s/2 = n_\mathrm{w} + n_\mathrm{u}$ in cases A, B and D, and $M = n_\mathrm{w}$ in case E where $N_s = 2n_\mathrm{w}$; with $n_\mathrm{w} = n_\mathrm{u} + n_\varphi + n_\mu$ from (41),

$$M = n_\mathrm{w} + n_\mathrm{u} = 2n_\mathrm{u} + n_\varphi + n_\mu = m \quad (\ell \neq 0), \qquad M = n_\mathrm{w} = n_\mathrm{u} + n_\varphi + n_\mu = m \quad (\ell = 0), \tag{56}$$

so the edge system is square in all cases and never over- or under-determined. At degree four ($n_\mathrm{u} = 15$, $n_\varphi = 3$, $n_\mu = 5$, $n_\mathrm{w} = 23$) this is $N_s = 76$, $m = M = 38$; at degree eighteen (57, 17, 19, 93) it is $N_s = 300$, $m = M = 150$; at the reported degree twenty-two (69, 21, 23, 113) it is $N_s = 364$, $m = M = 182$. The singular-value decomposition is still used, but as a square solve instead of a least-squares fit: the requirement is that the rank equal the column count at a cut-off of 10^{-12} of the largest singular value, and a deficient system is reported as such and is not regularized. Row and column scaling are applied as before, and the conditioning at the operating point is 1.4×10^2 after scaling.

A.5 Normalization of the modal power

The power (29) is bilinear in a mode and its conjugate, so the power carried by the j-th outgoing mode scales with the square of its excitation amplitude and with the normalization of its eigenvector. The eigenvector is normalized by $\boldsymbol{y}^{(j)} \to \boldsymbol{y}^{(j)}/\sqrt{\mathsf{y}^{\mathsf{H}}\mathsf{M}\mathsf{y}}$, which fixes its mass norm but leaves the *sign* of the power free: the physical power is

$$P_j = |a_j|^2\, \mathcal{P}_x\Big(\boldsymbol{y}^{(j)}\Big), \tag{57}$$

and the degree-invariance of P_j is the test that the normalization has been applied correctly. At the operating point, for the backward mode,

$$\begin{aligned} n = 18: &\quad \mathcal{P}_x = -0.0178923, \quad |a| = 1.03848, \quad P_B = -0.0192958, \\ n = 22: &\quad \mathcal{P}_x = -0.0154248, \quad |a| = 1.11846, \quad P_B = -0.0192957, \\ n = 26: &\quad \mathcal{P}_x = -0.0132455, \quad |a| = 1.20697, \quad P_B = -0.0192957, \\ n = 30: &\quad \mathcal{P}_x = -0.0113777, \quad |a| = 1.30227, \quad P_B = -0.0192956, \end{aligned} \tag{58}$$

in which the raw power falls by 36 % and the squared amplitude rises by 57 % across the four degrees while the physical power is constant to the six digits shown. That constancy is the numerical statement of (57) and it is why the reported share is degree-independent at the 10^{-6} level reported in section 4.3.

B Numerical resolution and convergence

Resolution. The reported solutions use polynomial degree $n = 22$, that is $N = 23$ Legendre modes per field ($N - 2 = 21$ for the grounded potential), and every integral is evaluated with $\max(64, 2n + 10)$ Gauss–Legendre nodes, which resolves the mass, stiffness and coupling matrices to machine precision at the degrees used here. The two electrical cases differ only in the potential space: the grounded basis is built from the combinations that vanish on both faces, and the open-circuit basis keeps all N modes, because a constant potential is not a gauge for a nonzero in-plane wavenumber. No field is discretized in x or in y.

Convergence and conservation. Between degree 18 and degree 30 both the operating-point share and the branch maximum change by less than 10^{-5}, and successive differences at the branch maximum decrease monotonically with degree. Refining the quadrature from 64 to 256 nodes at fixed basis changes no reported digit. The energy identity (30) is satisfied to 4×10^{-12} of the incident power at the reported solutions, and the partition at the branch maximum varies by 1.0×10^{-6} in absolute terms between degrees 14 and 34. These are the quantities that license quoting the shares to the precision used in the text.

References

[1] Zhong Lin Wang and Jinhui Song. Piezoelectric nanogenerators based on zinc oxide nanowire arrays. *Science*, 312(5771):242–246, 2006.

[2] Ü. Özgür, Ya. I. Alivov, C. Liu, A. Teke, M. A. Reshchikov, S. Doğan, V. Avrutin, S.-J. Cho, and H. Morkoç. A comprehensive review of ZnO materials and devices. *Journal of Applied Physics*, 98(4):041301, 2005.

[3] A. R. Hutson and Donald L. White. Elastic wave propagation in piezoelectric semiconductors. *Journal of Applied Physics*, 33(1):40–47, 1962.

[4] D. L. White. Amplification of ultrasonic waves in piezoelectric semiconductors. *Journal of Applied Physics*, 33(8):2547–2554, 1962.

[5] C. L. Zhang, X. Y. Wang, W. Q. Chen, and J. S. Yang. Propagation of extensional waves in a piezoelectric semiconductor rod. *AIP Advances*, 6(4):045301, 2016.

[6] L. L. Zhang, Z. Zhao, X. F. Hu, G. Q. Nie, and J. X. Liu. Exact multi-field coupling modeling and analysis of piezoelectric semiconductor plates. *Applied Mathematics and Mechanics (English Edition)*, 46(7):1331–1346, 2025.

[7] A. K. Tagantsev. Piezoelectricity and flexoelectricity in crystalline dielectrics. *Physical Review B*, 34(8):5883–5889, 1986.

[8] M. S. Majdoub, P. Sharma, and T. Çağin. Enhanced size-dependent piezoelectricity and elasticity in nanostructures due to the flexoelectric effect. *Physical Review B*, 77(12):125424, 2008.

[9] P. V. Yudin and A. K. Tagantsev. Fundamentals of flexoelectricity in solids. *Nanotechnology*, 24(43):432001, 2013.

[10] Pavlo Zubko, Gustau Catalan, and Alexander K. Tagantsev. Flexoelectric effect in solids. *Annual Review of Materials Research*, 43(1):387–421, 2013.

[11] S. P. Shen and S. L. Hu. A theory of flexoelectricity with surface effect for elastic dielectrics. *Journal of the Mechanics and Physics of Solids*, 58(5):665–677, 2010.

[12] U. K. Bhaskar, N. Banerjee, A. Abdollahi, E. Solanas, G. Rijnders, and G. Catalan. Flexoelectric MEMS: towards an electromechanical strain diode. *Nanoscale*, 8(3):1293–1298, 2016.

[13] R. D. Mindlin. Micro-structure in linear elasticity. *Archive for Rational Mechanics and Analysis*, 16(1):51–78, 1964.

[14] Y. Qu, F. Jin, and J. Yang. Effects of mechanical fields on mobile charges in a composite beam of flexoelectric dielectrics and semiconductors. *Journal of Applied Physics*, 127(19):194502, 2020.

[15] Y. L. Qu, G. Y. Zhang, X.-L. Gao, and F. Jin. A new model for thermally induced redistributions of free carriers in centrosymmetric flexoelectric semiconductor beams. *Mechanics of Materials*, 171:104328, 2022.

[16] Y. L. Qu, F. Jin, and J. S. Yang. Bending of a flexoelectric semiconductor plate. *Acta Mechanica Solida Sinica*, 35:434–445, 2022.

[17] Y. Qu, E. Pan, F. Zhu, and Q. Deng. A thermodynamically consistent theory for flexoelectronics: interaction between strain gradient and electric current in flexoelectric semiconductors. *International Journal of Engineering Science*, 208:104165, 2025.

[18] H. Hu, Z. Liu, Q. Liu, X. Jiang, X. Zhuang, and T. Rabczuk. Computational flexoelectronics: a framework for analyzing PN junctions in flexoelectric semiconductors. *International Journal of Mechanical Sciences*, 305:110740, 2025.

[19] W. J. Yang, Q. Deng, X. Liang, and S. P. Shen. Lamb wave propagation with flexoelectricity and strain gradient elasticity considered. *Smart Materials and Structures*, 27(8):085003, 2018.

[20] W. J. Yang, X. Liang, Q. Deng, and S. P. Shen. Rayleigh wave propagation in a homogeneous centrosymmetric flexoelectric half-space. *Ultrasonics*, 103:106105, 2020.

[21] Fengyu Jiao, Peijun Wei, and Yueqiu Li. Wave propagation through a flexoelectric piezoelectric slab sandwiched by two piezoelectric half-spaces. *Ultrasonics*, 82:217–232, 2018.

[22] Fengyu Jiao, Peijun Wei, Yahong Zhou, and Xiaoli Zhou. Wave propagation through a piezoelectric semiconductor slab sandwiched by two piezoelectric half-spaces. *European Journal of Mechanics – A/Solids*, 75:70–81, 2019.

[23] S. Mondal, S. A. Sahu, and S. Nirwal. Analysis of wave scattering on piezoelectric surface with flexoelectric effect. *Mathematical Methods in the Applied Sciences*, 45(12):7700–7714, 2022.

[24] M. Biswas and S. A. Sahu. On the Rayleigh wave velocity in n-type piezoelectric semiconductors with enhanced flexoelectricity. *Mathematics and Mechanics of Solids*, 30(12):2822–2842, 2025.

[25] Vipin Gupta, Fuad A. Awwad, Emad A. A. Ismail, Hijaz Ahmad, and Soumik Das. Piezo-semiconductor-void interplay in the size-dependent dynamics of Rayleigh waves within the Moore–Gibson–Thompson framework. *International Journal of Numerical Methods for Heat & Fluid Flow*, 35(9):3400–3428, 2025.

[26] M. G. Vavva, V. C. Protopappas, L. N. Gergidis, A. Charalambopoulos, D. I. Fotiadis, and D. Polyzos. Velocity dispersion of guided waves propagating in a free gradient elastic plate: application to cortical bone. *The Journal of the Acoustical Society of America*, 125(5):3414–3427, 2009.

[27] S. Sidhardh and M. C. Ray. Dispersion curves for Rayleigh–Lamb waves in a micro-plate considering strain gradient elasticity. *Wave Motion*, 86:91–109, 2019.

[28] Peter J. Torvik. Reflection of wave trains in semi-infinite plates. *The Journal of the Acoustical Society of America*, 41(2):346–353, 1967.

[29] B. A. Auld and E. M. Tsao. A variational analysis of edge resonance in a semi-infinite plate. *IEEE Transactions on Sonics and Ultrasonics*, 24(5):317–326, 1977.

[30] R. D. Gregory and I. Gladwell. The reflection of a symmetric Rayleigh–Lamb wave at the fixed or free edge of a plate. *Journal of Elasticity*, 13(2):185–206, 1983.

[31] Bruno Morvan, Nicolas Wilkie-Chancellier, Hugues Duflo, Alain Tinel, and Jean Duclos. Lamb wave reflection at the free edge of a plate. *The Journal of the Acoustical Society of America*, 113(3):1417–1425, 2003.

[32] S. Santhanam and R. Demirli. Reflection of Lamb waves obliquely incident on the free edge of a plate. *Ultrasonics*, 53(1):271–282, 2013.

[33] C. Prada, O. Balogun, and T. W. Murray. Laser-based ultrasonic generation and detection of zero-group velocity Lamb waves in thin plates. *Applied Physics Letters*, 87(19):194109, 2005.

[34] A. H. Meitzler. Backward-wave transmission of stress pulses in elastic cylinders and plates. *The Journal of the Acoustical Society of America*, 38(5):835–842, 1965.

[35] C. Prada, D. Clorennec, and D. Royer. Local vibration of an elastic plate and zero-group velocity Lamb modes. *The Journal of the Acoustical Society of America*, 124(1):203–212, 2008.

[36] M. Germano, A. Alippi, A. Bettucci, and G. Mancuso. Anomalous and negative reflection of Lamb waves in mode conversion. *Physical Review B*, 85(1):012102, 2012.

[37] B. Gérardin, J. Laurent, C. Prada, and A. Aubry. Negative reflection of Lamb waves at a free edge: tunable focusing and mimicking phase conjugation. *The Journal of the Acoustical Society of America*, 140(1):591–600, 2016.

[38] I. A. Veres, C. Grünsteidl, D. M. Stobbe, and T. W. Murray. Broad-angle negative reflection and focusing of elastic waves from a plate edge. *Physical Review B*, 93(17):174304, 2016.

[39] S. Bramhavar, C. Prada, A. A. Maznev, A. G. Every, T. B. Norris, and T. W. Murray. Negative refraction and focusing of elastic Lamb waves at an interface. *Physical Review B*, 83(1):014106, 2011.

[40] R. Zhu, X. N. Liu, G. K. Hu, C. T. Sun, and G. L. Huang. Negative refraction of elastic waves at the deep-subwavelength scale in a single-phase metamaterial. *Nature Communications*, 5(1):5510, 2014.

[41] A. T. I. Adamou and R. V. Craster. Spectral methods for modelling guided waves in elastic media. *The Journal of the Acoustical Society of America*, 116(3):1524–1535, 2004.

[42] Ivan Bartoli, Alessandro Marzani, Francesco Lanza di Scalea, and Erasmo Viola. Modeling wave propagation in damped waveguides of arbitrary cross-section. *Journal of Sound and Vibration*, 295(3–5):685–707, 2006.

[43] Jie Shen. Efficient spectral-Galerkin method I. Direct solvers of second- and fourth-order equations using Legendre polynomials. *SIAM Journal on Scientific Computing*, 15(6):1489–1505, 1994.

[44] M. Zhao, J. Niu, C. Lu, B. Wang, and C. Fan. Effects of flexoelectricity and strain gradient on bending vibration characteristics of piezoelectric semiconductor nanowires. *Journal of Applied Physics*, 129(16):164301, 2021.

[45] Yilin Qu, Feng Zhu, Ernian Pan, Feng Jin, and Hiroyuki Hirakata. Analysis of wave-particle drag effect in flexoelectric semiconductor plates via Mindlin method. *Applied Mathematical Modelling*, 118:541–555, 2023.

[46] Vipin Gupta, Sunita Kumawat, Arun Kumar Yadav, Taizo Maruyama, and Bandar Almohsen. Finite element analysis of second-order strain-gradient elasticity with ellipsoidal nonlocal characteristic lengths. *Finite Elements in Analysis and Design*, 263:104681, 2027.

[47] Dezhi Li, Shuangpeng Li, Chunli Zhang, and Weiqiu Chen. Propagation characteristics of shear horizontal waves in piezoelectric semiconductor nanoplates incorporating surface effect. *International Journal of Mechanical Sciences*, 247:108201, 2023.

[48] Mohammed Aldandani, Soumik Das, Arun Kumar Yadav, Murat Yaylacı, and Vipin Gupta. Dispersion and damping characteristics of Rayleigh-type waves in KG nonlocal fractional piezo-thermo-electric semiconductor medium. *International Journal of Numerical Methods for Heat & Fluid Flow*, 36(8):3021–3047, 2026.

[49] Vipin Gupta, Syed Modassir Hussain, Murat Yaylacı, and Soumik Das. Heat map analysis of wave dynamics in nonlocal Kelvin–Voigt piezo-semiconductors under memory-dependent heat flux. *Continuum Mechanics and Thermodynamics*, 38(3):41, 2026.